

Environmental Health

Clinical Approach to Total Toxic Load



ROBERT ROUNTREE, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Robert Rountree, MD, disclosed he was a consultant and stockholder with Thorne.

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Define the terms toxin, toxicant, total toxic load, xenobiotic, exposome, and biotransformation.
2. Recall five clinical patterns of an elevated total toxic load.
3. Explain how antecedents, triggers, mediators, and clinical findings provide evidence of an individual's susceptibility and exposure to toxic substances and the effects of total toxic load.
4. Explain the mechanisms of the phase I and phase II biotransformation processes and why it is important to initiate therapeutic interventions to keep them balanced.
5. Explain how dysfunction on other nodes of the Matrix may impair processes that decrease total toxic load and how the total toxic load may be a mediator of dysfunction on other nodes of the Matrix.

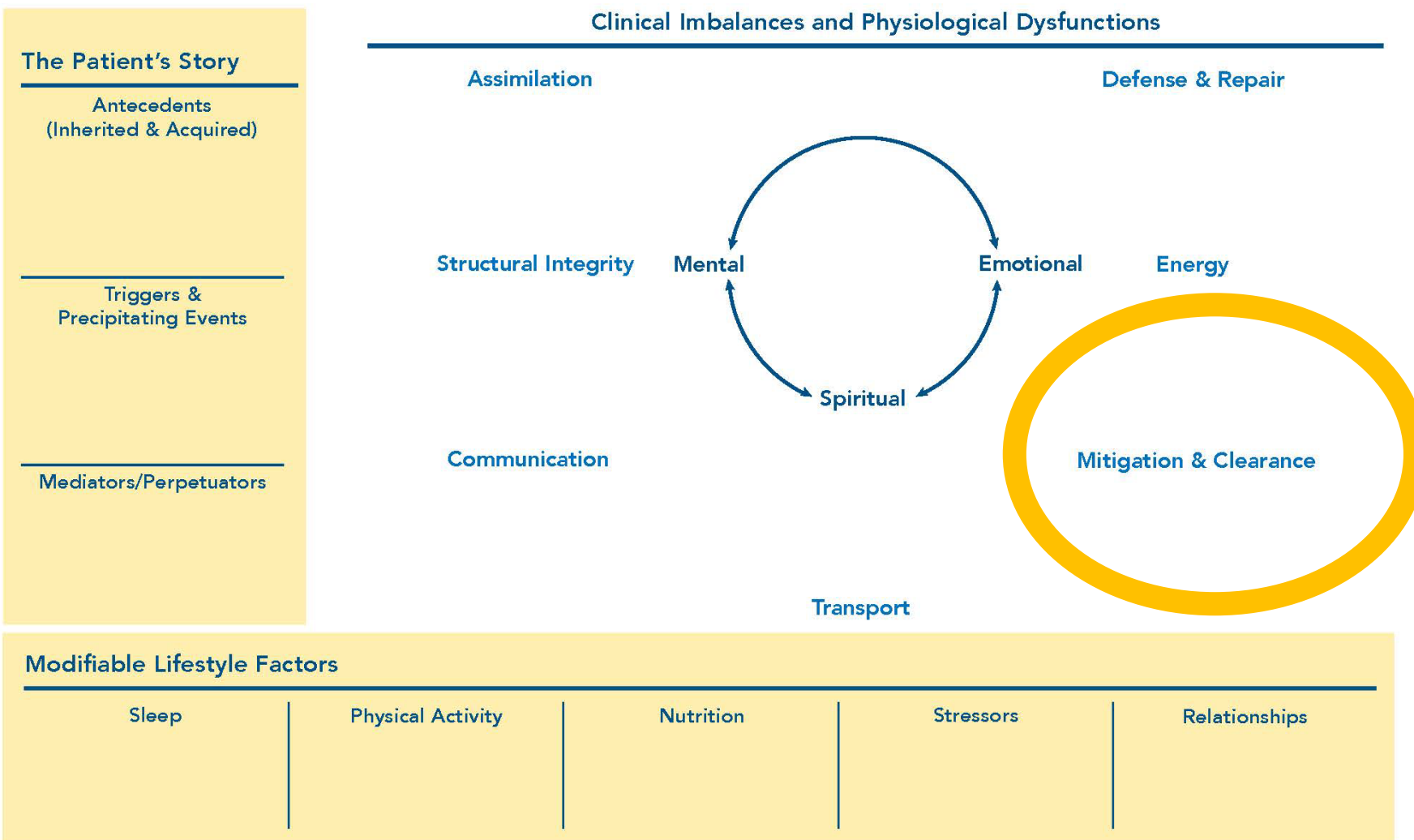
IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify clinical findings of the effects of elevated total toxic load and document them on the IFM Timeline and Matrix.
2. Identify and document the antecedents, triggers, and mediators of susceptibility and exposure to toxicants and toxins on the IFM Timeline and Matrix.
3. Explain to a patient how their antecedents, triggers, mediators, and clinical findings provide evidence of their susceptibility and exposure to toxins and toxicants and the effects of total toxic load.
4. Initiate interventions that address undernourishment, reduce toxicant exposures, and address gut health that are capable of safely and effectively reducing the total toxic load.



Functional Medicine Matrix



Name: _____ Date: _____ CC: _____

Clinical Focus

- Toxins & toxicants: terms, definitions, and common examples.



“What's out there making us sick?”

Contemporary scientific research in epigenetics, environmental health, and molecular medicine verifies what medical history has repeatedly and consistently confirmed—that deficiency and toxicity states contribute to disease.

Remediation of dysfunctional biochemistry and reduction of toxicant exposures has potential to deter chronic illness and restore health.

Foundational Terms and Definitions (Classical and Functional Toxicology)

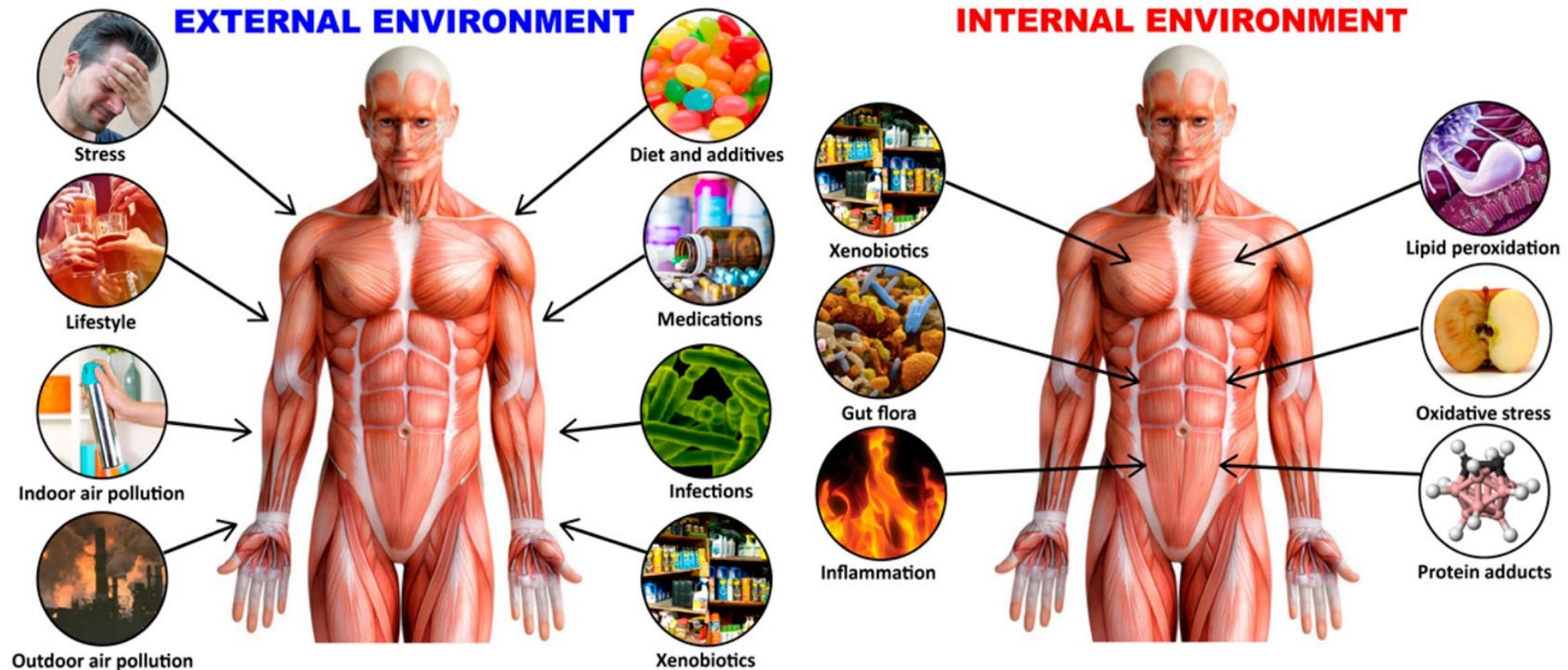
- **Toxin:** A toxic substance *produced naturally*, often by living organisms.
- **Toxicant:** A toxic agent or substance *produced by humans or introduced into the environment by human activities*.
- **Xenobiotic** (“foreign to life”): A compound found in the body that is not naturally produced or expected to be present under normal physiologic conditions; *toxicity implied but not inherent*.
- **Toxic Load/Body Burden:** The total quantity of toxins and toxicants found in the body at a given time. Reflects past exposures and ongoing accumulation.

Foundational Terms and Definitions

Exposome:

The combined and cumulative exposure to exogenous and internally produced *physiologically impactful* substances *from all sources throughout the lifespan*. This includes chemical, physical, biological, social, and lifestyle exposures.

The External and Internal Exposome



The **EXPOSOME**: An individual's lifetime exposure to a variety of external and internal factors. Collective investigation of the exposome is indicated.

Long Description: internal and external exposome

A diagram listing external and internal environmental exposome factors

1. External environment

- Stress
- Lifestyle
- Indoor air pollution
- Outdoor air pollution
- Diet and additives
- Medications
- Infections
- Xenobiotics

2. Internal environment

- Xenobiotics
- Gut flora
- Inflammation
- Lipid peroxidation
- Oxidative stress
- Protein adducts

Toxins: Common Examples

- **Mycotoxins:** aflatoxins, trichothecenes, gliotoxins, etc.
- **Plant toxins:** ricin (castor beans), cyanogenic glycosides (cassava), solanine (green potatoes), cardiac glycosides (digitoxin)
- **Bacteria:**
 - Exotoxins: secreted substances (e.g., botulinum toxins), cholera toxin, colibactin (genotoxic)
 - Endotoxins: part of bacterial membrane (LPS)
- **Aquatic biotoxins:** microcystins (fresh-water cyanobacteria), saxitoxin (marine), ciguatoxin (marine)

References

TOXINS – COMMON EXAMPLES

- Ghazaei C. Advances in the study of bacterial toxins, their roles and mechanisms in pathogenesis. *Malays J Med Sci*. 2022;29(1):4-17. doi:10.21315/mjms2022.29.1.2
- Shekhar R, Raghavendra VB, Rachitha P. A comprehensive review of mycotoxins, their toxicity, and innovative detoxification methods. *Toxicol Rep*. 2025;14:101952. doi:10.1016/j.toxrep.2025.101952
- Štefanac T, Grgas D, Landeka Dragičević T. Xenobiotics-division and methods of detection: a review. *J Xenobiot*. 2021;11(4):130-141. doi:10.3390/jox11040009
- Vilariño N, Louzao MC, Abal P, et al. Human poisoning from marine toxins: unknowns for optimal consumer protection. *Toxins (Basel)*. 2018;10(8):324. doi:10.3390/toxins10080324
- World Health Organization. Natural toxins in food. Published March 10, 2023. Accessed May 9, 2025. <https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food>

Xenobiotics: Common Examples

- Toxic metals (Pb, Hg, As, Cd)
- Organochlorines (DDT, PCBs, dioxins)
- Flame retardants (PBDEs)
- Organophosphates
- PFAS/PFOS (“forever chemicals”)
- Glyphosate
- Neonicotinoids (aka neonics)
- Microplastics and nanoplastics
- Pharmaceuticals
- Phytochemicals (e.g., polyphenols)

Clinical Focus

- Clinical pattern recognition of elevated total toxic load.

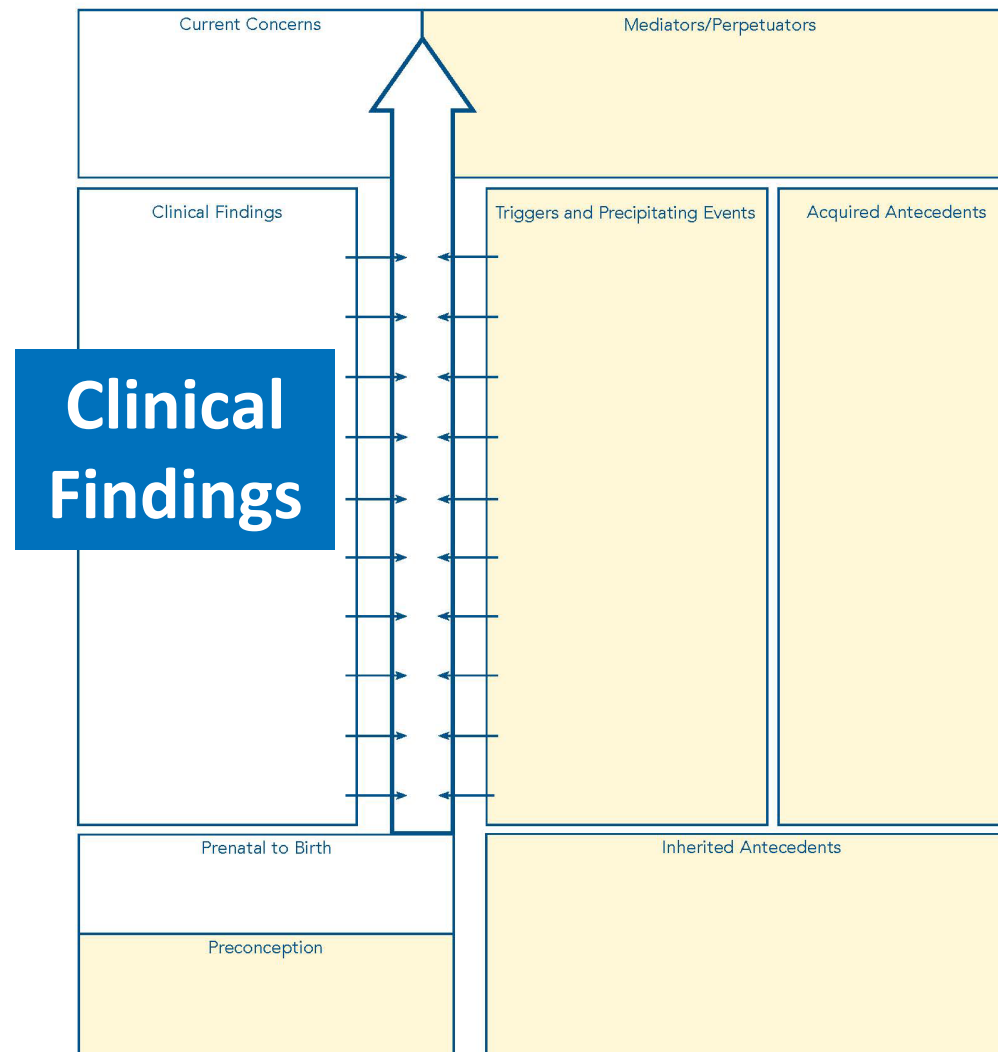


Gather:

What patterns of clinical findings suggest toxic substances as root causes?



Functional Medicine Timeline



Name: _____ Date: _____ CC: _____



Louis Pasteur

*“Dans les champs de l’observation
le hasard ne favorise que
les esprits préparés.”*

(In the fields of observation, chance favors
only the prepared mind).

Long Description: Quote

- A quote by the 1800s French scientist Louis Pasteur. “In the fields of observation, chance favors only the prepared mind.”

Environmental Chemicals: Association With Chronic Disease in Children

- Over the past half-century, non-communicable disease rates in children have risen sharply.
- A large body of evidence links multiple pediatric noncommunicable diseases to synthetic chemicals.
- Toxic chemicals cross the placenta, and children are more vulnerable to the effects of toxic chemicals than adults.
- Widespread childhood exposure to toxic chemicals can damage the health, economic viability, and security of an entire society.



Pesticides and Children's Health Outcomes

Children are more susceptible to the health effects of pesticides due to lower body mass and higher metabolic rate.

Conditions with increased risks include:

- Neurodevelopmental
- Endocrinological
- Immunological
- Cancer

Chronic Conditions and Functional Limitations in Children on the Rise

The prevalence of US youth living with a chronic condition or functional limitation has risen over the past two decades from around one in five to nearly one in three.

Associations noted:

- Prediabetes
- Asthma
- Hypertension



Recognizable Clinical Findings of Toxic Effect:

1. Neurologic & Psychiatric Toxicity
2. Immunologic Toxicity
3. Metabolic/Mitochondrial Toxicity
4. Endocrine Disruption
5. Genotoxicity/Carcinogenesis

Clinical Findings of the Five Patterns of Toxicity



1. NEUROLOGIC & PSYCHIATRIC MANIFESTATIONS

- Concentration, memory, & learning issues
- ADHD, autism spectrum disorders
- Chronic headaches, insomnia
- Peripheral neuropathy, tremor
- Autonomic dysfunction
- Neurodegenerative diseases:
Alzheimer's, Parkinson's, ALS
- Mood disorders:
depression, anxiety, irritability, etc.

Stress Reduction Interventions:

Reducing Toxic Load to Support Mitigation of Anxiety

A study that used NHANES data from 2007 to 2012 to investigate toxic metals such as cadmium, lead, and mercury and their relationship to anxiety disorders demonstrated a positive association between blood cadmium levels and anxiety.

Exposure to multiple heavy metals (cadmium, lead, and mercury) was positively associated with anxiety risk, with cadmium contributing the most to the overall effect.

Clinical Findings of the Five Patterns of Toxicity

2. IMMUNE DYSFUNCTION

- Atopic syndrome: allergies, asthma
- Autoimmune diseases
- Chronic inflammatory diseases
- Recurrent or chronic infections
- Chronic dermatitis (e.g., chloracne)
- Adverse food reactions (AFR)
- Multiple chemical sensitivities (MCS): TILT

Clinical Findings of the Five Patterns of Toxicity

3. MITOCHONDRIAL & METABOLIC

- Obesity—abnormal weight gain
- Metabolic syndrome; type 2 diabetes mellitus
- Atherosclerosis
- Fatigue; chronic fatigue syndrome
- Fibromyalgia, myalgias, weakness
- Osteoporosis

Clinical Findings of the Five Patterns of Toxicity

4. ENDOCRINE DISRUPTION

- Hormonal abnormalities
- Hypothalamic-pituitary-adrenal (HPA axis)
- Thyroid dysfunction
- Metabolic syndrome/T2DM
- Female: premature puberty, PMS, PCOS, infertility, endometriosis, fibroids
- Male: oligospermia, sperm dysmotility

Clinical Findings of the Five Patterns of Toxicity

5. GENOTOXICITY/CARCINOGENESIS

- Cancer
- Developmental disorders/birth defects
- Miscarriages

Environmental Health: Part 2

Clinical Approach to Total Toxic Load

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

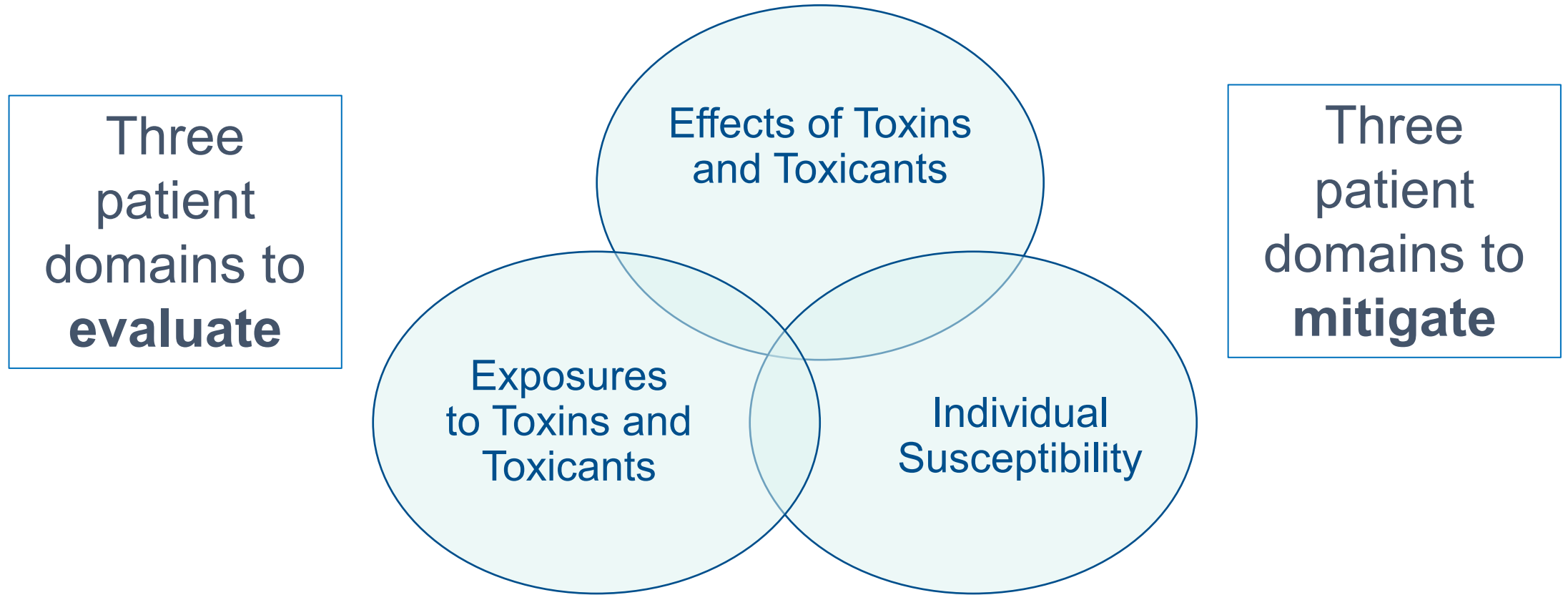
Clinical Focus

- Foundational concepts related to total toxic load:
 - Exposure
 - Susceptibility
 - Effect



The “Toxic Triad”

Susceptibility – Exposure – Effect



Environmental Chemicals: Increasing Exposure

- An estimated 350,000 manufactured chemicals, chemical mixtures, and plastics are currently listed in global inventories. Most are produced from fossil fuels – gas, oil, and coal.
- Production has expanded 50-fold since 1950, is currently increasing by about 3% per year, and is projected to triple by 2050.
- Environmental pollution and human exposure are widespread.

Functional Medicine & Toxicology: Fundamental Principles

Exposure to environmental toxins and toxicants is widespread, increasing, and lifelong.

Susceptibility is largely determined by an individual's unique ability to mitigate, biotransform, or metabolize and eliminate (or “clear”) exogenous and endogenous toxins and toxicants.

The effects of these toxic substances are dependent on the “total body burden” because small amounts of multiple toxic substances can act synergistically to impair physiological functions.

Functional Medicine & Toxicology: Fundamental Principles

- Physiological dysfunctions mediated by toxins and toxicants can lead to chronic disease in susceptible individuals.
- Diet, lifestyle, and gut health interventions can minimize exposure to and increase the metabolism, biotransformation, & clearance/elimination of toxic substances.
- Nutraceuticals and other supplements may provide added benefits to minimize the effects of toxic exposure and enhance clearance.

Disease Risk
From Toxic
Substances
(End Result)



Toxic Potency



Cumulative
Exposure



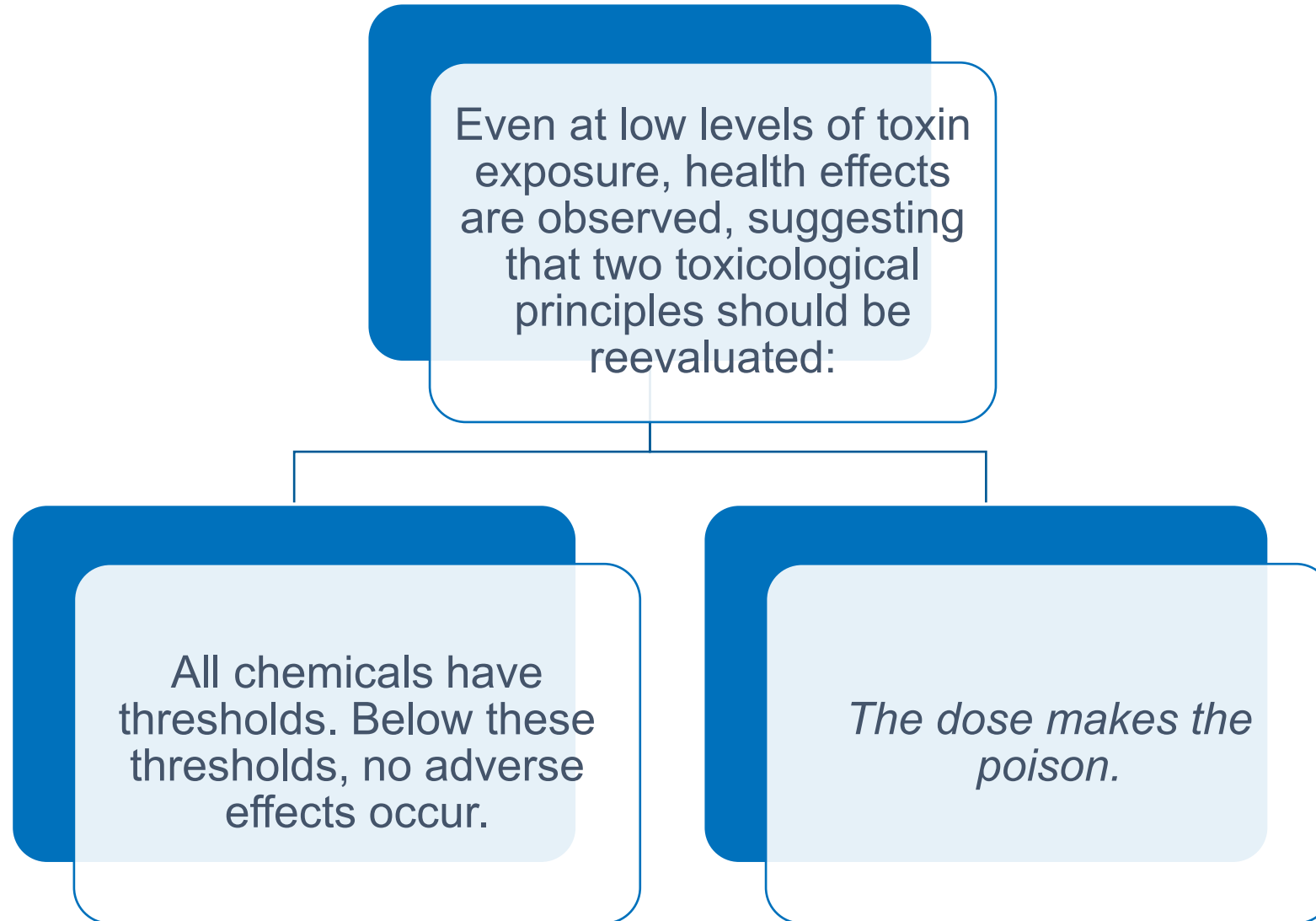
Susceptibility

Foundational Concept of Classical Toxicology

"All things are poison, and nothing is without poison; the dose alone makes a thing not poison."

– Paracelsus (1493-1541)





Disease and Mortality Risks Attributable to Environmental Exposures

In 2016, 13.7 million deaths, representing 24% of global fatalities, were linked to environmental factors. This includes a significant portion of deaths related to air pollution.

About 25-33% of the global burden of disease can be attributed to environmental risk factors.

- World Health Organization. SDG target 3.9 mortality from environmental pollution. Accessed May 9, 2025. https://www.who.int/data/gho/data/themes/topics/sdg-target-3_9-mortality-from-environmental-pollution
- World Health Organization. Environmental health. Accessed May 9, 2025. https://www.who.int/health-topics/environmental-health#tab=tab_2

Functional Medicine & Toxicology: Fundamental Principles

Hormones and other endogenously produced chemicals commonly exert physiologically potent effects at levels of parts per million (ppm) or parts per trillion (ppt).

Consequently, “low” levels of xenobiotics or toxicants (nanograms or picograms per mL = ppm/ppt) can be biologically significant.

Reproductive Hormone Reference Ranges – Serum

Reproductive Hormone (serum)	Female (adult)	Male (adult)
Estrone	10-60 pg/mL	
Estradiol		20-50 pg/mL
• Estradiol – follicular phase	100-300 pg/mL	
• Estradiol – mid-cycle peak	40-200 pg/mL	
• Estradiol – luteal	<10 pg/mL	
Follicle-stimulating hormone (FSH)		1-7 mIU/mL (1-7 U/L)
• FSH – follicular/luteal	2-9 mIU/mL (2-9 U/L)	
• FSH – mid-cycle peak	4-22 mIU/mL (4-22 U/L)	
• FSH – postmenopausal	>30 mIU/mL (>30 U/L)	

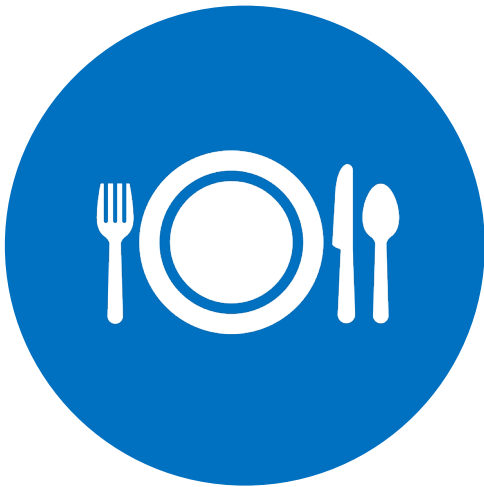
Reproductive Hormone Reference Ranges – Serum

Reproductive Hormone (serum)	Female (adult)	Male (adult)
Progesterone		0.12-0.3 ng/mL
• Female – follicular	• 0.02-0.9 ng/mL	
• Female – luteal	• 2-30 ng/mL	
Luteinizing hormone (LH)		2-9 mIU/mL (2-9 U/L)
• Female – follicular/luteal 1-12 mIU/mL (1-12 U/L)		
• Female – mid-cycle peak 9-80 mIU/mL (9-80 U/L)		
• Female – postmenopausal >30 mIU/mL (>30 U/L)		
Testosterone	18-54 ng/dL	291-1,100 ng/dL

State of the Science: Endocrine-Disrupting Chemicals (EDCs)

- Many endocrine-related disorders are on the rise.
- The speed which the increases in disease incidence have occurred in recent decades rules out genetic factors as the sole plausible explanation.
- Numerous laboratory studies support the idea that chemical exposures contribute to endocrine disorders in humans and wildlife.
- The most sensitive window of exposure to EDCs is during critical periods of development, such as fetal development & puberty.

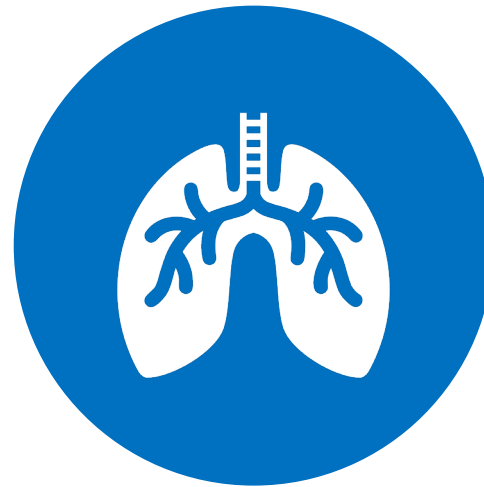
Toxic Exposures: We are what we...



EAT



DRINK



BREATHE



TOUCH

Disease Risk
From Toxic
Substances
(End Result)



Toxic Potency



Cumulative
Exposure



Susceptibility

Long Description: Disease Risk

- Disease risk from toxic substances, the end result, is expressed as a mathematical equation
- Disease risk from toxic substances equals Toxic Potency multiplied by Cumulative Exposure multiplied by Susceptibility
- On this slide, Cumulative Exposure is emphasized

Total Toxic Load Is Determined by:

Total Exposure
(to Toxic
Substances)



Ability to
Mitigate,
Biotransform, or
Metabolize



Clearance or
Elimination



Source: George McCaa, US Bureau of Mines

Disease Risk
From Toxic
Substances
(End Result)



Toxic Potency



Cumulative
Exposure

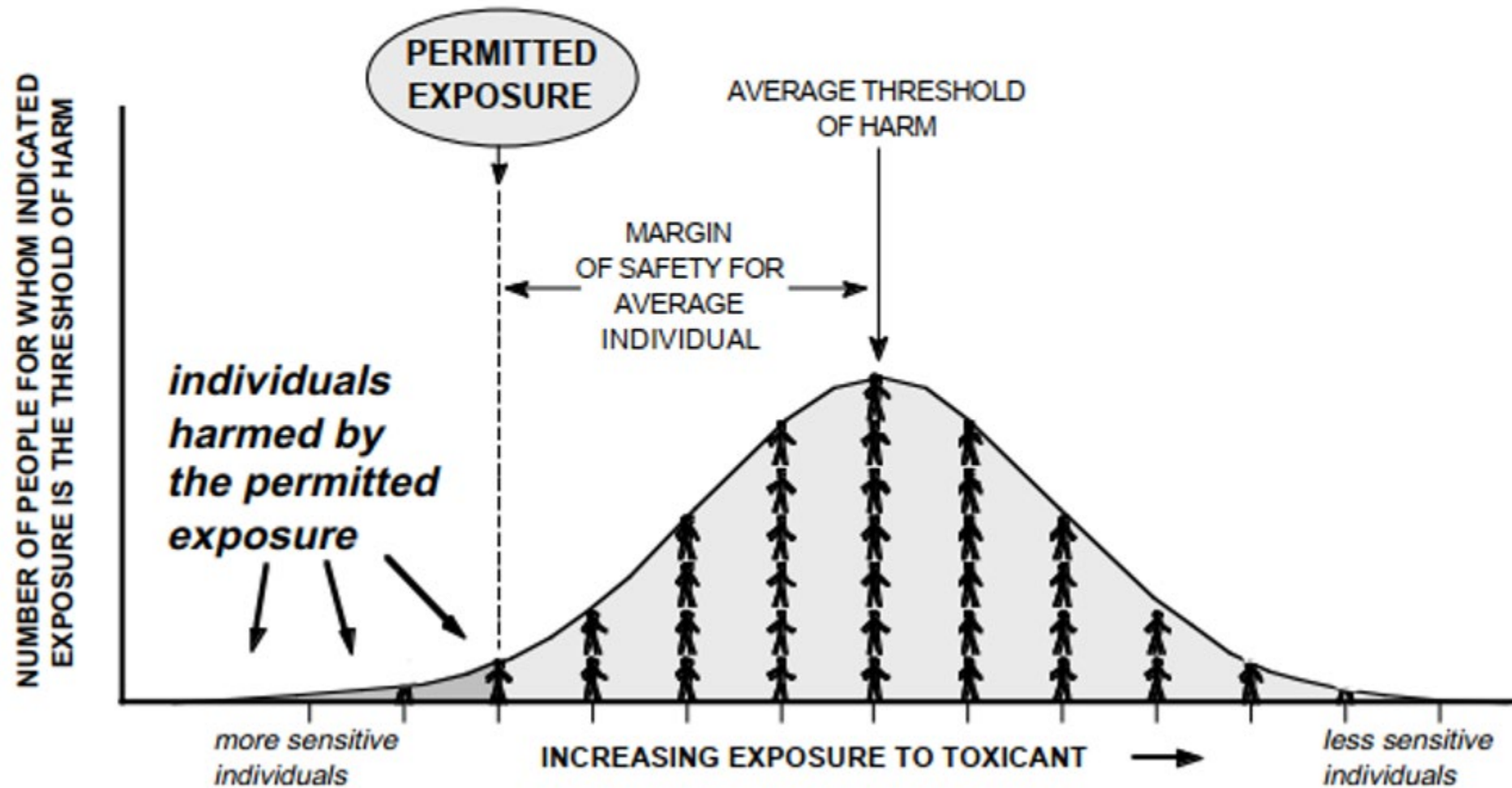


Susceptibility

Long Description: Disease Risk

- Disease risk from toxic substances, the end result, is expressed as a mathematical equation
- Disease risk from toxic substances equals Toxic Potency multiplied by Cumulative Exposure multiplied by Susceptibility
- On this slide, Susceptibility is emphasized

Spectrum of Vulnerability



Long Description: Spectrum of Vulnerability

Diagram titled Spectrum of Vulnerability.

This graphic shows how individuals vary in their sensitivity to a toxic substance, forming a bell-shaped (normal) distribution curve. The horizontal axis represents increasing exposure to a toxicant, ranging from more sensitive individuals on the left to less sensitive individuals on the right. The vertical axis represents the number of people for whom a given exposure level is the threshold of harm.

A vertical dashed line labeled "**Permitted Exposure**" is placed to the left of the peak of the curve. This indicates a regulatory limit intended to be safe. However, a shaded area under the curve to the left of this line represents "**individuals harmed by the permitted exposure**"—those who are more sensitive than average.

The peak of the curve is labeled "**Average Threshold of Harm,**" showing where most people would begin to experience harm. The space between the permitted exposure and the average threshold of harm is labeled the "**Margin of Safety for Average Individual.**"

- The overall message is that while exposure limits are set to protect the general population, they may not protect more sensitive individuals, who can still be harmed at legally permitted levels of exposure.

Environmental Health: Part 3

Clinical Approach to Total Toxic Load

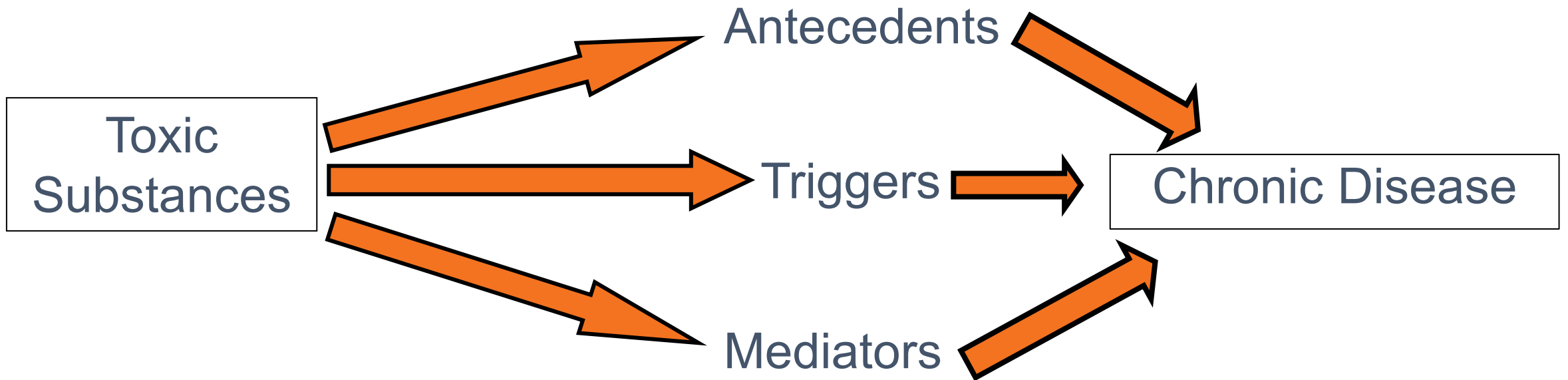
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Antecedents, triggers, and mediators of elevated total toxic load.



Functional Toxicology



Inherited Antecedents

Elevated Total Toxic Load

- Family history of patterns and clinical findings
- Genetic susceptibilities (SNPs)
- Maternal and paternal(?) body burden
- Maternal gestational exposures

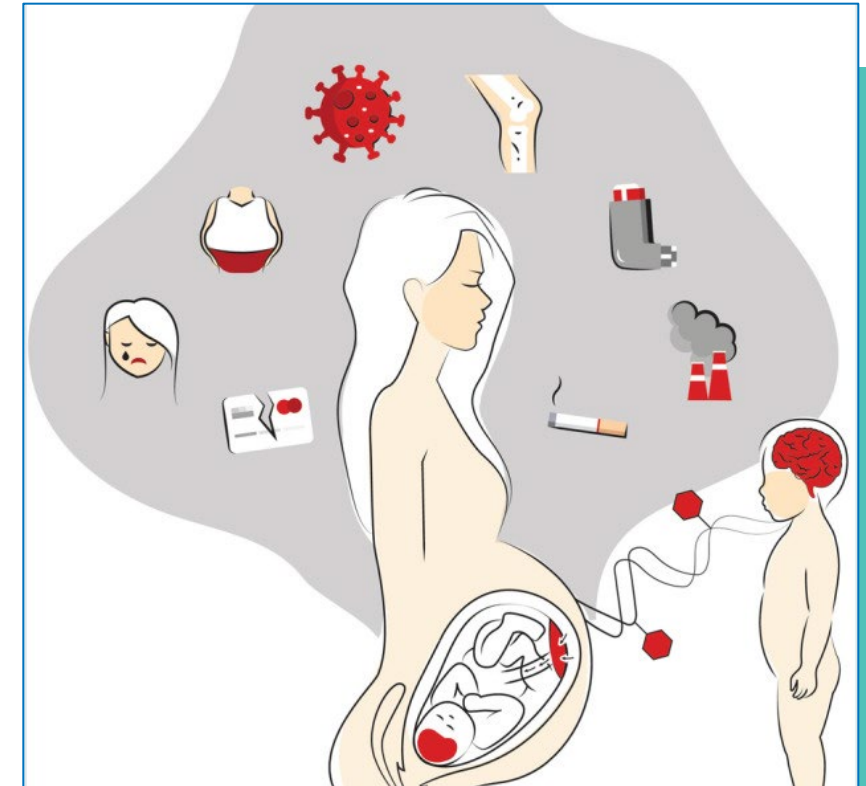
Inherited Antecedents: Associations With Autism Spectrum Disorder

Maternal exposure to environmental pollutants associated with ASD:

- Air pollution
- Toxic metals
- Pesticides
- Endocrine-disrupting chemicals

Putative mechanisms of action:

- Increased permeability of blood-brain barrier during the gestational period and infancy
- Interference with normal gene activity



Acquired Antecedents: Elevated Total Toxic Load

- Childhood exposures
- Occupational exposures
- Residential exposures
- Medical/dental exposures
- Previous dietary patterns
- Previous lifestyle behaviors

Acquired Antecedents: Elevated Toxic Load From Childhood Exposures

In a longitudinal birth cohort study (n=9,065) that investigated association of air and noise pollution exposure in pregnancy:

Outcomes in youth (13-24 yrs old):

- Higher exposure to fine particulate matter (PM_{2.5}):
 - Associated with increased psychotic experiences in pregnancy and childhood
 - Higher rates of depression in pregnancy
- Higher noise pollution exposure in childhood and adolescence:
 - Associated with increased anxiety



Triggers (Single Exposures): Elevated Total Toxic Load

- Childhood exposures
- Occupational exposures
- Residential exposures
- Medical/dental exposures
- Weight loss (crash diets)
- Other known single significant exposures

Triggers:

Elevated Total Toxic Load Due to Rapid Weight Loss

- Xenobiotics (including POPs) are stored in adipose
- Rapid weight loss releases POPs into bloodstream
- Study of 94 obese adolescents:
 - Lost 4 to 42 kg during a five-month diet program
 - POPs increased by 1-3.5% per kg weight loss



Mediators: Elevated Total Toxic Load

- Adverse lifestyle behaviors
- Undernutrition/dehydration
- Constipation/dysbiosis
- Recreational and leisure activities
- Sick building syndrome
- Liver disease/damage
- Transcriptional effects on biotransformation enzymes
- Other ongoing exposures

Mediators: Increased Toxic Load Due to Environmental Exposures

Case-control study – proximity to golf course:

- Greater risk of Parkinson's disease (PD) if within 1-3 miles of a golf course.
- If drinking water source in proximity to golf course, nearly double the risk of PD.



Elevated Total Toxic Load

Lifestyle Factor Mediators

Modifiable Lifestyle Factors

Sleep

- Disturbed sleep
- New/toxic bedroom furnishings
- Poor air quality

Physical Activity

- Sedentarism
- Over-exercising
- Rapid weight loss
- Anhidrosis

Nutrition

- Ultra-processed foods
- Artificial sweeteners
- Fish and seafood
- Non-organic

Stressors

- Catecholamines
- Stress eating
- Financial stress
- Unsupported caregiving

Relationships

- Isolation
- Toxic relationships: negativity, abuse, codependence, mental illness, etc.



Long Description: Elevated Total Toxic Load Lifestyle Factor Mediators

- A depiction of the bottom section of the functional medicine matrix, listing modifiable lifestyle factors
- The modifiable lifestyle mediators that contribute to elevated total toxic load are:

1. Sleep factors

- Disturbed sleep
- New/toxic bedroom furnishings
- Poor air quality

2. Physical activity

- Sedentarism
- Over-exercising
- Rapid weight loss
- Anhidrosis

3. Nutrition

- Ultra-processed foods
- Artificial sweeteners
- Fish and seafood
- Non-organic

4. Stressors

- Catecholamines
- Stress eating
- Financial stress
- Unsupported caregiving

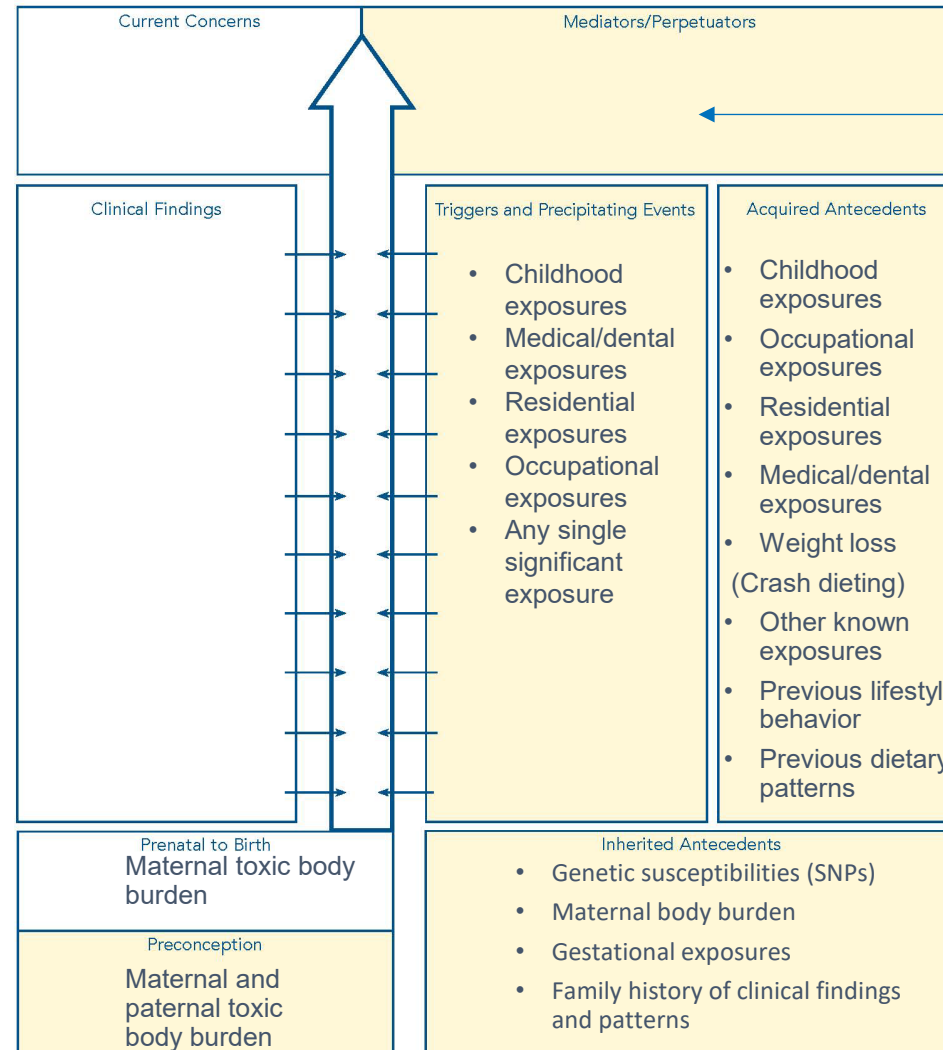
5. Relationships

- Isolation
- Toxic relationships: negativity, abuse, codependence, mental illness, etc.

Antecedents, Triggers, and Mediators of Elevated Total Toxic Load



Functional Medicine Timeline



- Lifestyle behaviors/patterns
- Undernutrition
- Dehydration
- Constipation
- Recreational toxins
- Sick building syndrome
- Any ongoing toxic exposure
- Liver disease/damage
- Transcriptional effects
- Genetic SNPs

Name: _____ Date: _____ CC: _____

Long Description: Antecedents, Triggers, and mediators of elevated total toxic load

- The functional medicine timeline showing antecedents, triggers, and mediators of elevated total toxic load

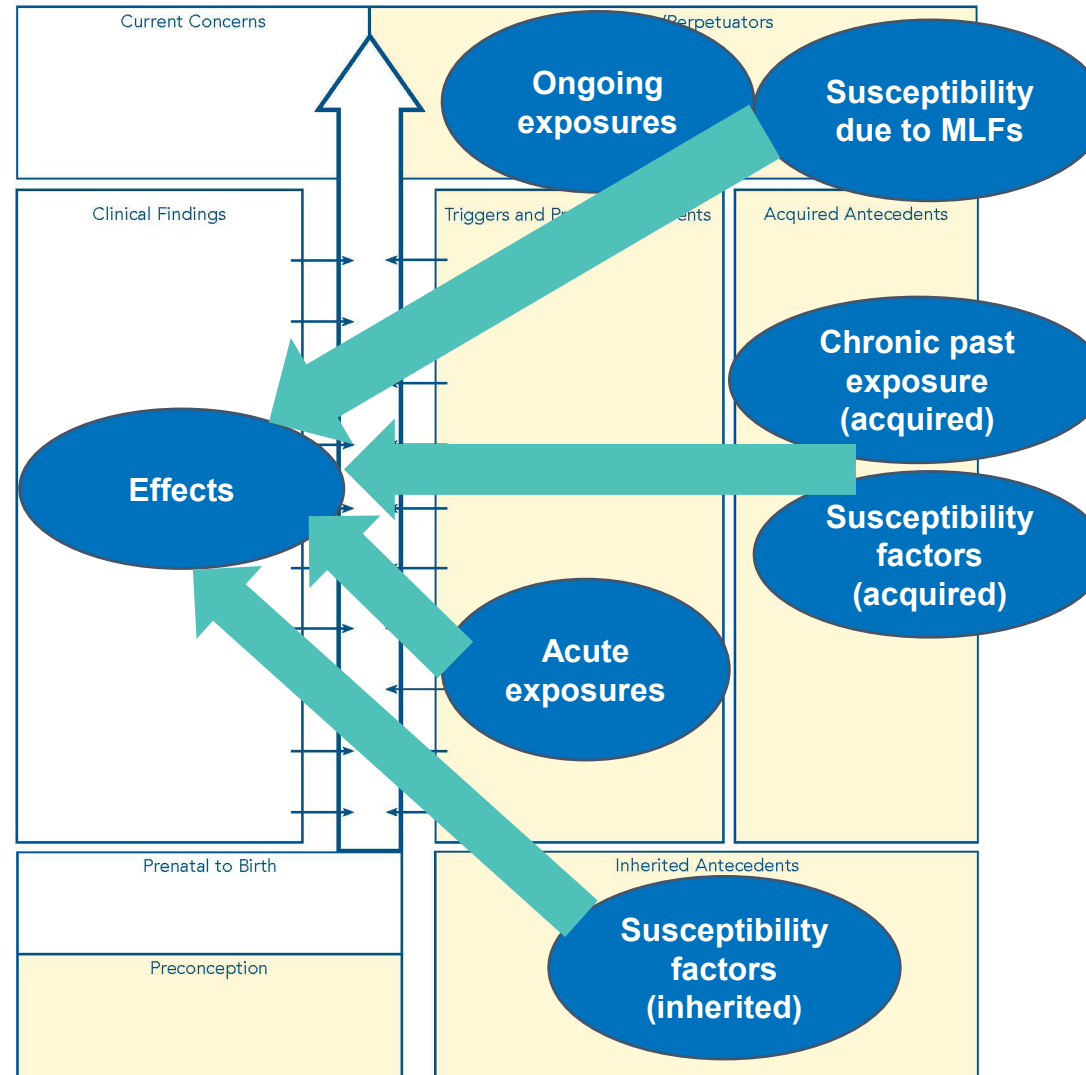
1. **Preconception** section includes
 - Maternal and paternal toxic body burden
2. **Prenatal to Birth** section includes
 - Maternal toxic body burden
3. **Inherited Antecedents** section includes
 - Genetic susceptibilities (SNPs)
 - Maternal body burden
 - Gestational exposures
 - Family history of clinical findings and patterns
4. **Triggers and Precipitating Events** section includes
 - Childhood exposures
 - Medical/dental exposures
 - Residential exposures
 - Occupational exposures
 - Any single significant exposure

5. **Acquired Antecedents** section includes
 - Childhood exposures
 - Occupational exposures
 - Residential exposures
 - Medical/dental exposures
 - Weight loss (Crash dieting)
 - Other known exposures
 - Previous lifestyle behavior
 - Previous dietary patterns
6. **Mediators/Perpetuators** section includes
 - Lifestyle behaviors/patterns
 - Undernutrition
 - Dehydration
 - Constipation
 - Recreational toxins
 - Sick building syndrome
 - Any ongoing toxic exposure
 - Liver disease/damage
 - Transcriptional effects
 - Genetic SNPs

Correlation of antecedents, triggers, and mediators with susceptibility, exposure, and effects of toxic substances.



Functional Medicine Timeline



Name: _____ Date: _____ CC: _____

Long Description: Correlation of antecedents, triggers, and mediators with susceptibility, exposure, and effects of toxic substances.

1. A functional medicine timeline illustrates how various exposures and susceptibility factors over a person's life contribute to health effects.
2. In the following sections of the timeline, these are some of the factors that ultimately contribute to health effects
 - In Inherited antecedents, consider Inherited Susceptibility Factors
 - In Triggers, consider Acute Exposures
 - In Acquired Antecedents, consider acquired Susceptibility Factors and Chronic Past Exposures
 - In Mediators and Perpetuators, consider Ongoing Exposures and Susceptibility to modifiable lifestyle factors.
3. These inputs represent how inherited traits, past and present toxic exposures, and modifiable lifestyle-related susceptibilities all converge to create effects that result in Clinical Findings.

Clinical Focus

- Clinical point-of-care-tools to assist evaluation of exposures to and effects of toxins and toxicants.



Point-of-Care Screening Questionnaires



Toxin Exposure Questionnaire

Patient Name _____ Date _____

Please mark your response for each of the following questions. Your provider will discuss your answers with you.

Food and Water	
1. Do you eat conventionally-farmed (non-organic) or genetically-modified fruits and vegetables?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
2. Do you eat conventionally-raised (non-organic) animal products (e.g., meat, poultry, dairy, eggs)?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
3. Do you eat canned or farmed fish and seafood?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
4. Do you eat processed foods (e.g., foods with added artificial colors, flavors, preservatives), deep-fried foods, or fast foods?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
5. Do you drink water from a well, spring, or cistern, or from plumbing pipes or fixtures installed before 1986?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
6. Do you drink sodas, juices, or other beverages with natural or refined sweeteners (high-fructose corn syrup, cane sugar, agave nectar, stevia, undiluted fruit juice, etc.) or artificial sweeteners (i.e., Equal® or aspartame; Sweet'N Low®, Sugar Twin®, or saccharin; Splenda® or sucralose; Sunett®, Sweet One®, or acesulfame-K; and neotame)?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No

Home and Work Environment	
1. Do you live in an apartment or home built before 1978 or in a mobile home, boat, or recreational vehicle (RV)?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
2. Does your home or workplace contain new furniture, bedding, or construction materials (paint, laminate flooring, etc.)?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
3. Does your home or workplace show signs of mold or water damage (e.g., cracking paint, ceiling leaks, decaying insulation or foam, visible mold, or damp areas in windows, crawlspaces, or the basement)?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
4. Are you exposed to toxic substances (e.g., treated lumber; lead paint, paint chips, or dust; broken mercury thermometers or fluorescent bulbs) at home or work?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
5. Are you exposed to conventional cleaning chemicals, disinfectants, hand sanitizers, air fresheners, scented candles, or other scented products at home or work?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No
6. Do you live or work near an industrial pollution source (e.g., highway, factory, incinerator, gas station, power plant)?	☐ Yes ☐ Sometimes ☐ In the Past ☐ No

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Toxin Exposure Questionnaire (TEQ)

Medical Symptoms Questionnaire (MSQ)

Patient Name _____ Date _____

Instructions
Use this point scale to rate each of the following symptoms based on your typical health profile for the past 14 days:
0 = Never or almost never have the symptom
1 = Occasionally have it; effect is not severe
2 = Occasionally have it; effect is severe
3 = Frequently have it; effect is not severe
4 = Frequently have it; effect is severe

Head
_____ Headaches
_____ Faintness
_____ Dizziness
_____ Insomnia
_____ Total

Eyes
_____ Watery or Itchy eyes
_____ Swollen, reddened, or sticky eyelids
_____ Bags or dark circles under eyes
_____ Blurred or tunnel vision (doesn't include nearsightedness or farsightedness)
_____ Total

Ears
_____ Itchy ears
_____ Earaches, ear infections
_____ Drainage from ear
_____ Ringing in ears, hearing loss
_____ Total

Nose
_____ Stuffy nose
_____ Sinus problems
_____ Hay fever
_____ Sneezing attacks
_____ Excessive mucus formation
_____ Total

Please continue on the next page

Version 3

Medical Symptoms Questionnaire (MSQ)

Toxin Exposure Questionnaire



Toxin Exposure Questionnaire

Patient Name _____ Date _____

Please mark your response for each of the following questions. Your provider will discuss your answers with you.

Food and Water

1. Do you eat conventionally-farmed (non-organic) or genetically-modified fruits and vegetables?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
2. Do you eat conventionally-raised (non-organic) animal products (e.g., meat, poultry, dairy, eggs)?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
3. Do you eat canned or farmed fish and seafood?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
4. Do you eat processed foods (e.g., foods with added artificial colors, flavors, preservatives), deep-fried foods, or fast foods?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
5. Do you drink water from a well, spring, or cistern, or from plumbing pipes or fixtures installed before 1986?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
6. Do you drink sodas, juices, or other beverages with natural or refined sweeteners (high-fructose corn syrup, cane sugar, agave nectar, stevia, undiluted fruit juice, etc.) or artificial sweeteners (i.e., Equal® or aspartame; Sweet'N Low®, Sugar Twin®, or saccharin; Splenda® or sucralose; Sunett®, Sweet One®, or acesulfame-K; and neotame)?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No

Home and Work Environment

1. Do you live in an apartment or home built before 1978 or in a mobile home, boat, or recreational vehicle (RV)?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
2. Does your home or workplace contain new furniture, bedding, or construction materials (paint, laminate flooring, etc.)?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
3. Does your home or workplace show signs of mold or water damage (e.g., cracking paint, ceiling leaks, decaying insulation or foam, visible mold, or damp areas in windows, crawlspaces, or the basement)?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
4. Are you exposed to toxic substances (e.g., treated lumber; lead paint, paint chips, or dust; broken mercury thermometers or fluorescent bulbs) at home or work?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
5. Are you exposed to conventional cleaning chemicals, disinfectants, hand sanitizers, air fresheners, scented candles, or other scented products at home or work?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No
6. Do you live or work near an industrial pollution source (e.g., highway, factory, incinerator, gas station, power plant)?
☐ Yes ☐ Sometimes ☐ In the Past ☐ No

Exposure Sources:

- Food and Water
- Home and Work Environment
- Travel and Recreation
- Medical and Personal Care

MSQ: Quantifies Symptom Severity

71 questions

Scored: 0-4

Total: 0-284

Accessible version in your toolkit.

**Medical Symptoms
Questionnaire (MSQ)**

Patient Name _____ Date _____

Instructions
Use this point scale to rate each of the following symptoms based on your typical health profile for the past 14 days:
0 = Never or almost never have the symptom 3 = Frequently have it; effect is not severe
1 = Occasionally have it; effect is not severe 4 = Frequently have it; effect is severe
2 = Occasionally have it; effect is severe

Head

____ Headaches
____ Faintness
____ Dizziness
____ Insomnia
____ Total

Eyes

____ Watery or itchy eyes
____ Swollen, reddened, or sticky eyelids
____ Bags or dark circles under eyes
____ Blurred or tunnel vision (doesn't include nearsightedness or farsightedness)
____ Total

Ears

____ Itchy ears
____ Earaches, ear infections
____ Drainage from ear
____ Ringing in ears, hearing loss
____ Total

Nose

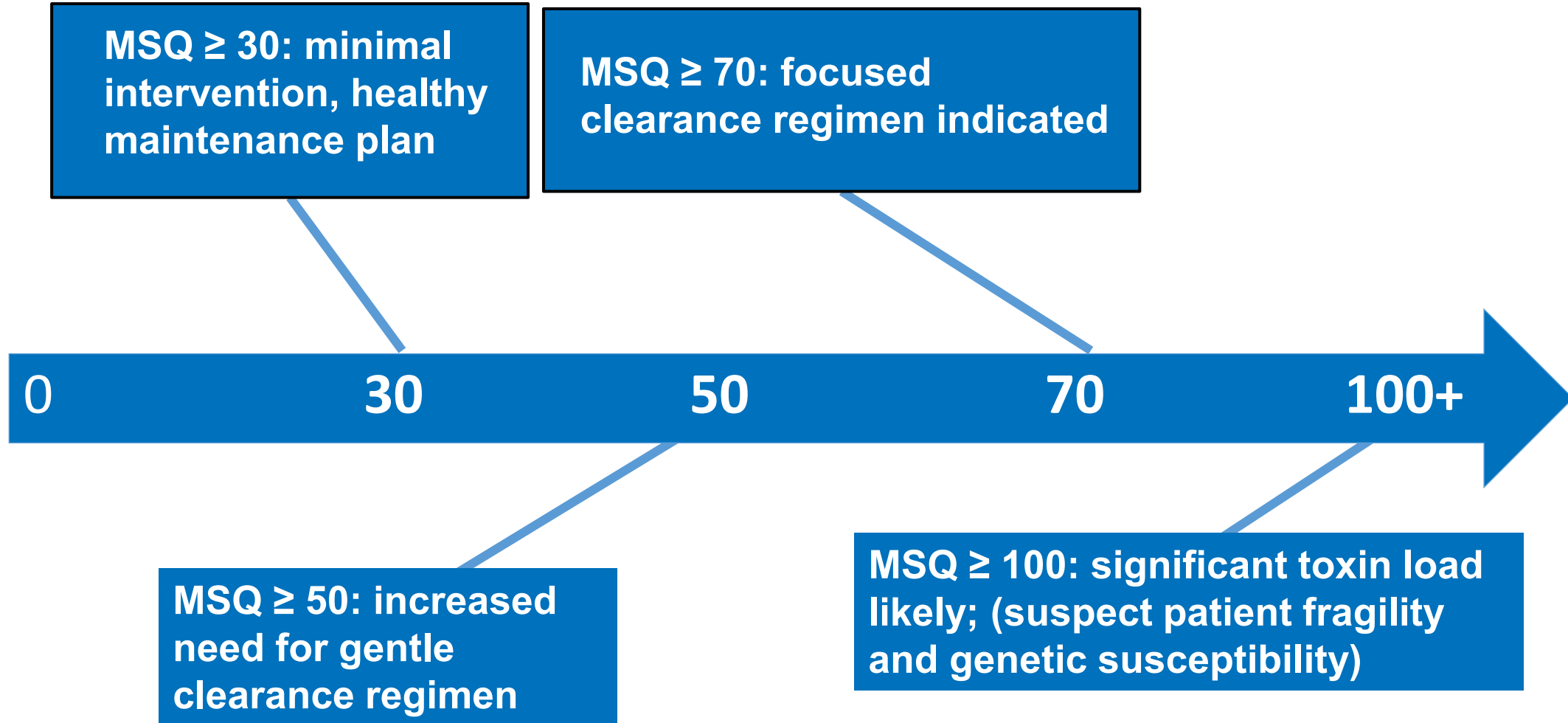
____ Stuffy nose
____ Sinus problems
____ Hay fever
____ Sneezing attacks
____ Excessive mucus formation
____ Total

Please continue on the next page

Version 3

1. HEAD
2. EYES
3. EARS
4. NOSE
5. MOUTH/THROAT
6. SKIN
7. HEART
8. LUNGS
9. DIGESTIVE TRACT
10. JOINTS/MUSCLE
11. WEIGHT
12. ENERGY/ACTIVITY
13. MIND
14. EMOTIONS
15. OTHER

MSQ: Quantifies Symptom Severity



Note: The Medical Symptoms Questionnaire (MSQ) is an unvalidated questionnaire.

Long Description: MSQ quantifies symptom severity

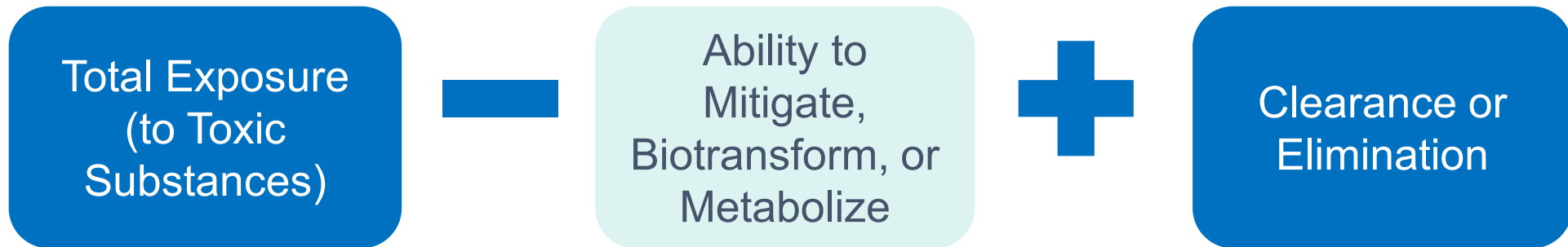
- A graphic explains how the MSQ (Medical Symptoms Questionnaire) score quantifies symptom severity and guides intervention strategies. It presents a horizontal arrow with key score thresholds:
 - **MSQ $\geq$ 30:** Suggests minimal intervention is needed and a healthy maintenance plan may be appropriate.
 - **MSQ $\geq$ 50:** Indicates an increased need for a gentle clearance regimen.
 - **MSQ $\geq$ 70:** A focused clearance regimen is indicated.
 - **MSQ $\geq$ 100:** Suggests a significant toxin load is likely, with concern for patient fragility and genetic susceptibility.
- This scale helps practitioners determine the intensity of therapeutic intervention based on a patient's symptom burden.

Clinical Focus

- Foundational concepts related to mitigation of total toxic load.



Total Toxic Load Is Determined by:



Multiple Modes of Mitigation:

Elimination is not the only physiological strategy

- Direct neutralization (e.g., via antioxidants or molecular moieties)
- Fibers and binders (e.g., charcoal)
- Sequestration/compartmentalization
- Enzymatic biotransformation

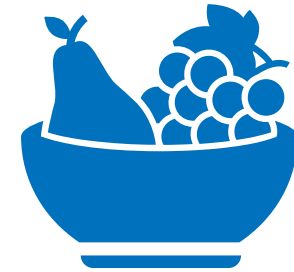
Mitigation and Neutralization: Antioxidants From Food



Oxidative stress caused by environmental toxins disrupts cellular antioxidant systems, inducing damage to cellular membrane lipids, proteins, and DNA.



Antioxidant nutrients and bioactive phytonutrients found in fruits and vegetables can support the neutralization of the effects of some toxins.



These nutrients include vitamins C, E, D, and polyphenols.

References

MITIGATION AND NEUTRALIZATION: ANTIOXIDANTS FROM FOOD

- Poljšak B, Fink R. The protective role of antioxidants in the defence against ROS/RNS-mediated environmental pollution. *Oxid Med Cell Longev*. 2014;2014:671539. doi:10.1155/2014/671539
- Rana A, Samtiya M, Dhewa T, Mishra V, Aluko RE. Health benefits of polyphenols: a concise review. *J Food Biochem*. 2022;46(10):e14264. doi:10.1111/jfbc.14264
- Schichlein KD, Smith GJ, Jaspers I. Protective effects of inhaled antioxidants against air pollution-induced pathological responses. *Respir Res*. 2023;24(1):187. doi:10.1186/s12931-023-02490-7
- Żwieretko W, Maruszewska A, Skórka-Majewicz M, et al. The influence of polyphenols on metabolic disorders caused by compounds released from plastics - review. *Chemosphere*. 2020;240:124901. doi:10.1016/j.chemosphere.2019.124901

Sequestration and Neutralization: Non-Enzymatic Detoxification Pathways

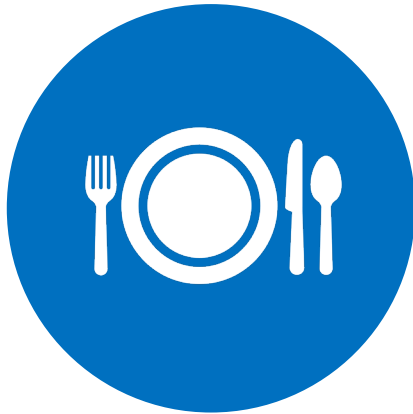
Intracellular non-enzymatic antioxidants such as glutathione (GSH) can efficiently mitigate cadmium-induced oxidative stress and toxicity.

Metallothioneins (MT) can effectively bind to toxic metals and scavenge ROS, protecting against Cd-induced oxidative stress.

Toxin Sequestration in the Body

- Toxins are stored in various locations throughout the body, including adipose tissue, bone liver, and kidneys.
 - Adipose tissue sequesters lipophilic toxins such as persistent organic pollutants (POPs).
 - About 94% of lead is absorbed by bones in adults.
- Bile also plays a role in sequestering and removing toxins from the body and is a common route of excretion for POPs.
 - Enterohepatic recirculation of many of these toxins can slow down their removal from the body. Use of binders or fiber from the diet can help sequester these toxins in the gut.

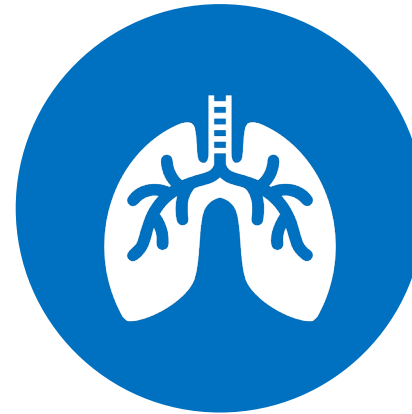
Toxic Exposures: We are what we...



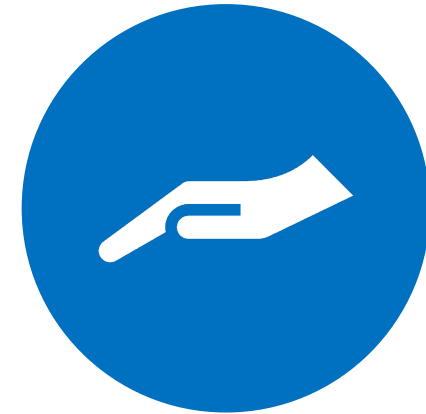
EAT



DRINK



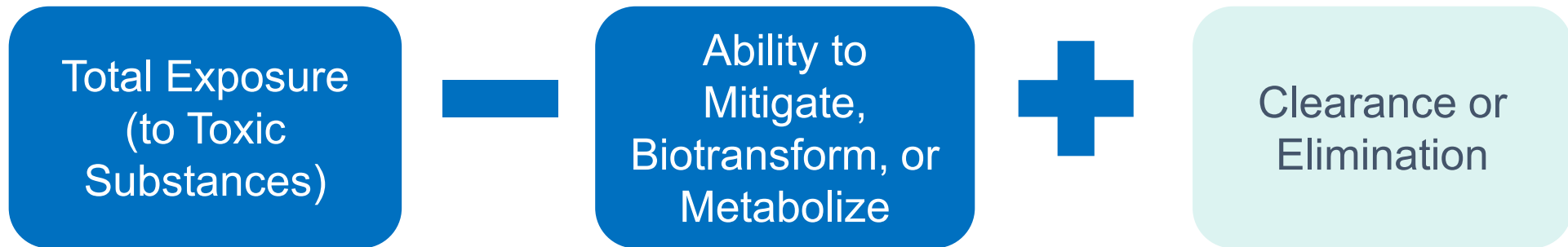
BREATHE



TOUCH

..and can't mitigate or clear.

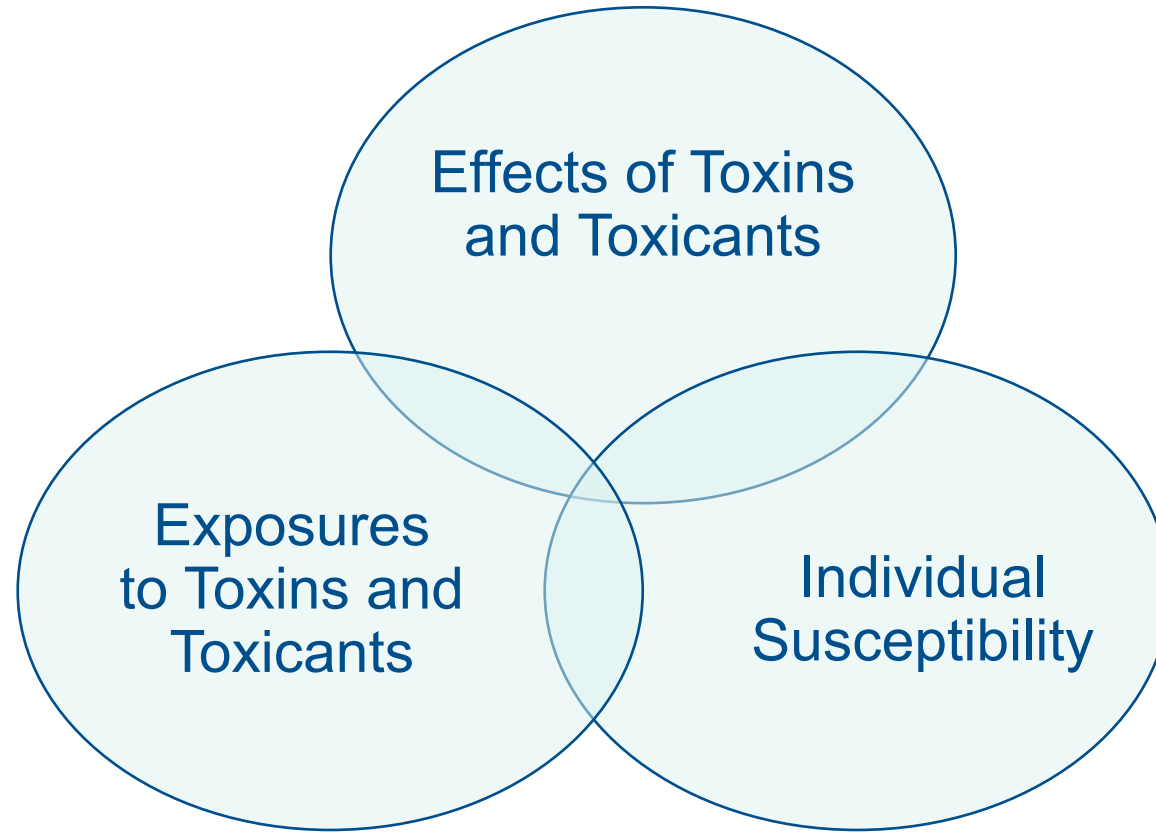
Total Toxic Load Is Determined by:



The “Toxic Triad”

Susceptibility – Exposure – Effect

Three
patient
domains to
evaluate



Three
patient
domains to
mitigate

Environmental Health: Part 4

Clinical Approach to Total Toxic Load

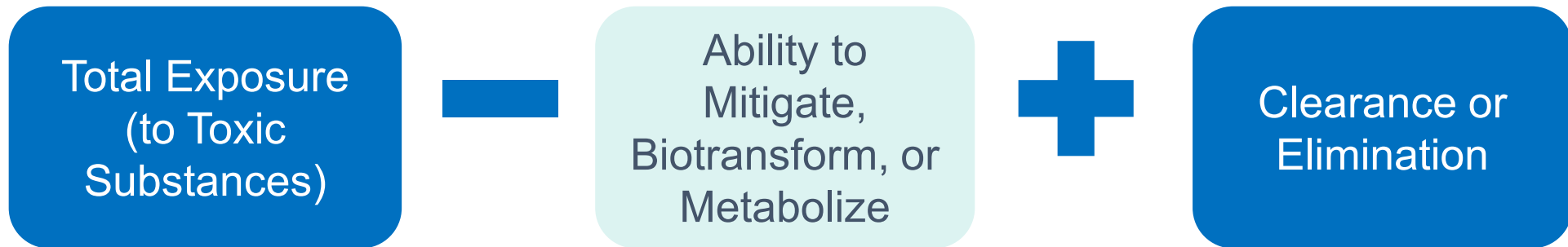
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

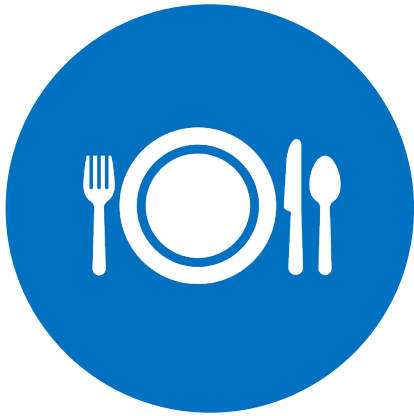
- Overview of biotransformation and clearance of toxins.



Total Toxic Load Is Determined by:



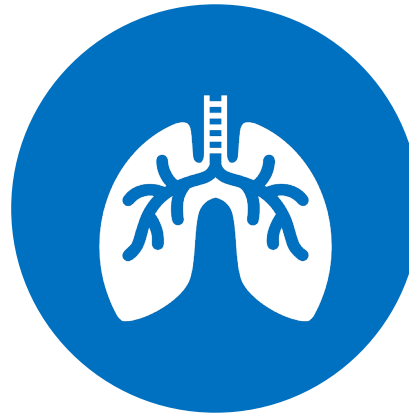
Toxic Exposures: We are what we...



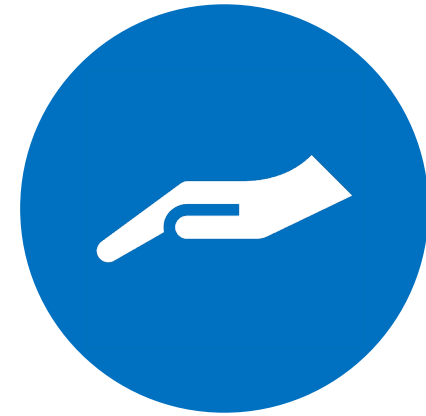
EAT



DRINK



BREATHE



TOUCH

...and can't mitigate, biotransform, or metabolize.

Detoxification

***More accurately described as detoxication
(may include clearance or biotransformation)***

Enzymatically driven biochemical alterations of a xenobiotic, phytochemical, or endogenous compound that render it less toxic and/or more readily cleared/excreted.

Detoxication Overview:

Xenobiotics, Phytochemicals, and Endogenous Compounds

Hydrophilic Compounds:
readily cleared/excreted in
urine and/or bile.

Lipophilic Compounds:
must be chemically altered
into hydrophilic compounds to
facilitate elimination.

Biotransformation is...

- A constantly active process
- ATP (energy) dependent
- Highly nutrient dependent
- Polymorphic: marked interindividual variability

Biotransformation of Lipophilic Toxicants

Biotransformation typically involves two steps:

- Phase I bioactivation (i.e., functionalization)
- Phase II conjugation

Biotransformation of Lipophilic Toxicants: Anatomical Locations

All tissues have some activity:

Greatest activity occurs
in liver.

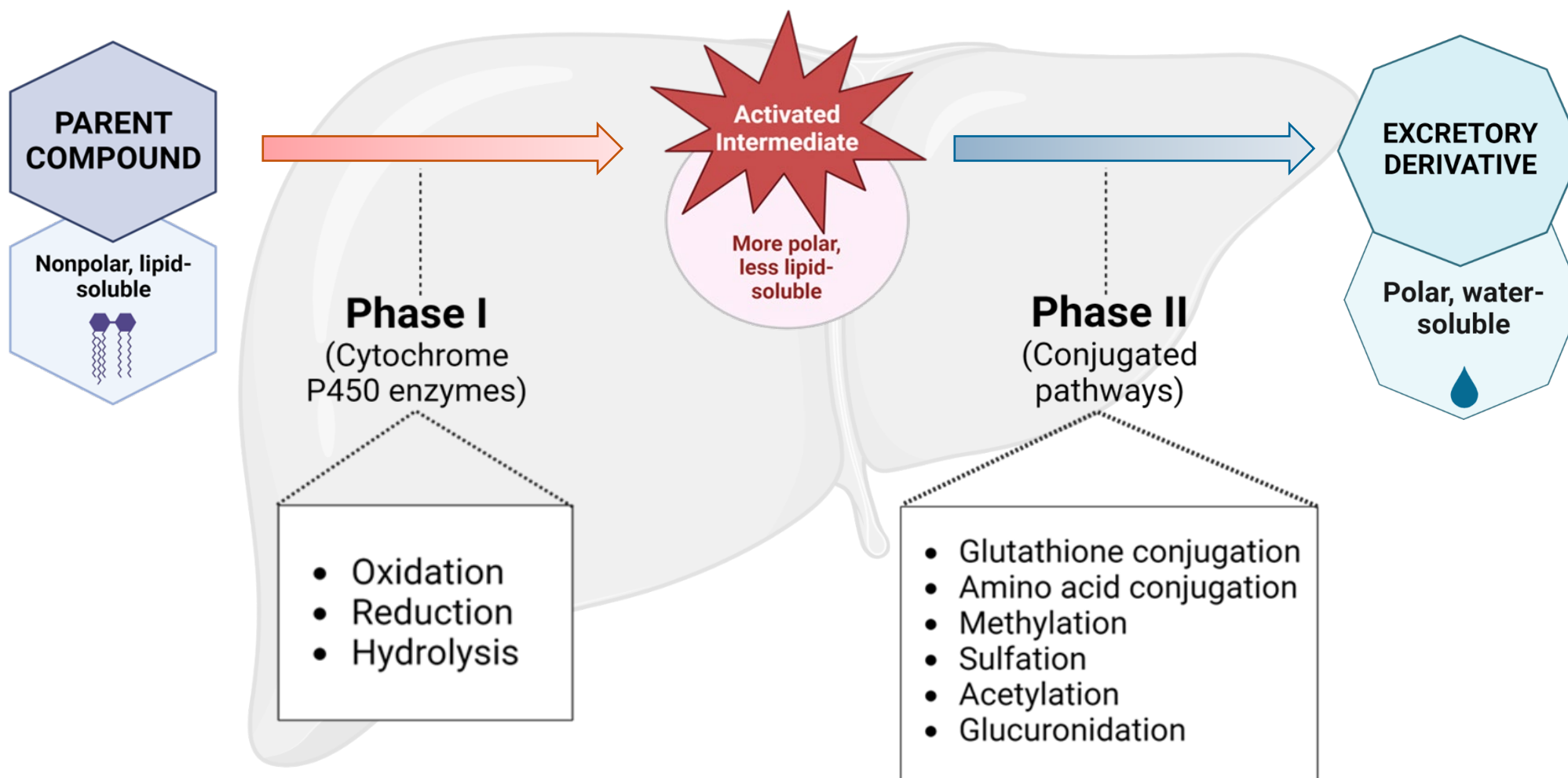
Gut microbiome also
plays significant role.

Relatively high activity in
GI tract, lungs, kidneys,
skin, brain, and blood-
brain barrier.

References: Biotransformation of Lipophilic Toxicants

- Agúndez JA, Jiménez-Jiménez FJ, Alonso-Navarro H, García-Martín E. Drug and xenobiotic biotransformation in the blood-brain barrier: a neglected issue. *Front Cell Neurosci*. 2014;8:335. Published 2014 Oct 17. doi:10.3389/fncel.2014.00335
- Enlo-Scott Z, Bäckström E, Mudway I, Forbes B. Drug metabolism in the lungs: opportunities for optimising inhaled medicines. *Expert Opin Drug Metab Toxicol*. 2021;17(5):611-625. doi:10.1080/17425255.2021.1908262
- Mega A, Marzi L, Kob M, Piccin A, Floreani A. Food and Nutrition in the Pathogenesis of Liver Damage. *Nutrients*. 2021;13(4):1326. Published 2021 Apr 16. doi:10.3390/nu13041326
- Phang-Lyn S, Llerena VA. Biochemistry, Biotransformation. [Updated 2023 Aug 14]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK544353/>
- Serenade M, Saedisomeolia A, Shekarabi M, Khorshidi M, Emami MR, Müller DJ. The effect of obesity, macronutrients, fasting and nutritional status on drug-metabolizing cytochrome P450s: a systematic review of current evidence on human studies. *Eur J Nutr*. 2021 Sep;60(6):2905-2921. doi: 10.1007/s00394-020-02421-y. Epub 2020 Nov 3. PMID: 33141242.
- Zarezadeh M, Saedisomeolia A, Shekarabi M, Khorshidi M, Emami MR, Müller DJ. The effect of obesity, macronutrients, fasting and nutritional status on drug-metabolizing cytochrome P450s: a systematic review of current evidence on human studies. *Eur J Nutr*. 2021 Sep;60(6):2905-2921. doi: 10.1007/s00394-020-02421-y. Epub 2020 Nov 3. PMID: 33141242.

Biotransformation: Essentials of the Process



Long Description: Biotransformation Essentials of the process

- A diagram illustrating the two-phase process of liver detoxification that converts nonpolar, lipid-soluble compounds into polar, water-soluble forms for excretion.
- **Starting Point – Parent Compound:** The process begins with a nonpolar, lipid-soluble compound that enters the liver.
- **Phase I (Cytochrome P450 enzymes):** This phase involves chemical reactions—oxidation, reduction, and hydrolysis—that convert the parent compound into a more polar, but potentially reactive and toxic, **activated intermediate**.
- **Activated Intermediate:** This intermediate is more polar and less lipid-soluble than the original compound but is not yet ready for excretion.
- **Phase II (Conjugation Pathways):** The activated intermediate undergoes further modification through conjugation reactions, including:
 - Glutathione conjugation
 - Amino acid conjugation
 - Methylation
 - Sulfation
 - Acetylation
 - Glucuronidation
- **End Point – Excretory Derivative:** The result is a polar, water-soluble compound that can be safely excreted from the body.
- **Summary:** Liver detoxification occurs in two phases—Phase I (modification) and Phase II (conjugation)—to make toxins water-soluble for elimination.

Phase I Reactions

Involve CYP450 enzymes in the smooth endoplasmic reticulum.

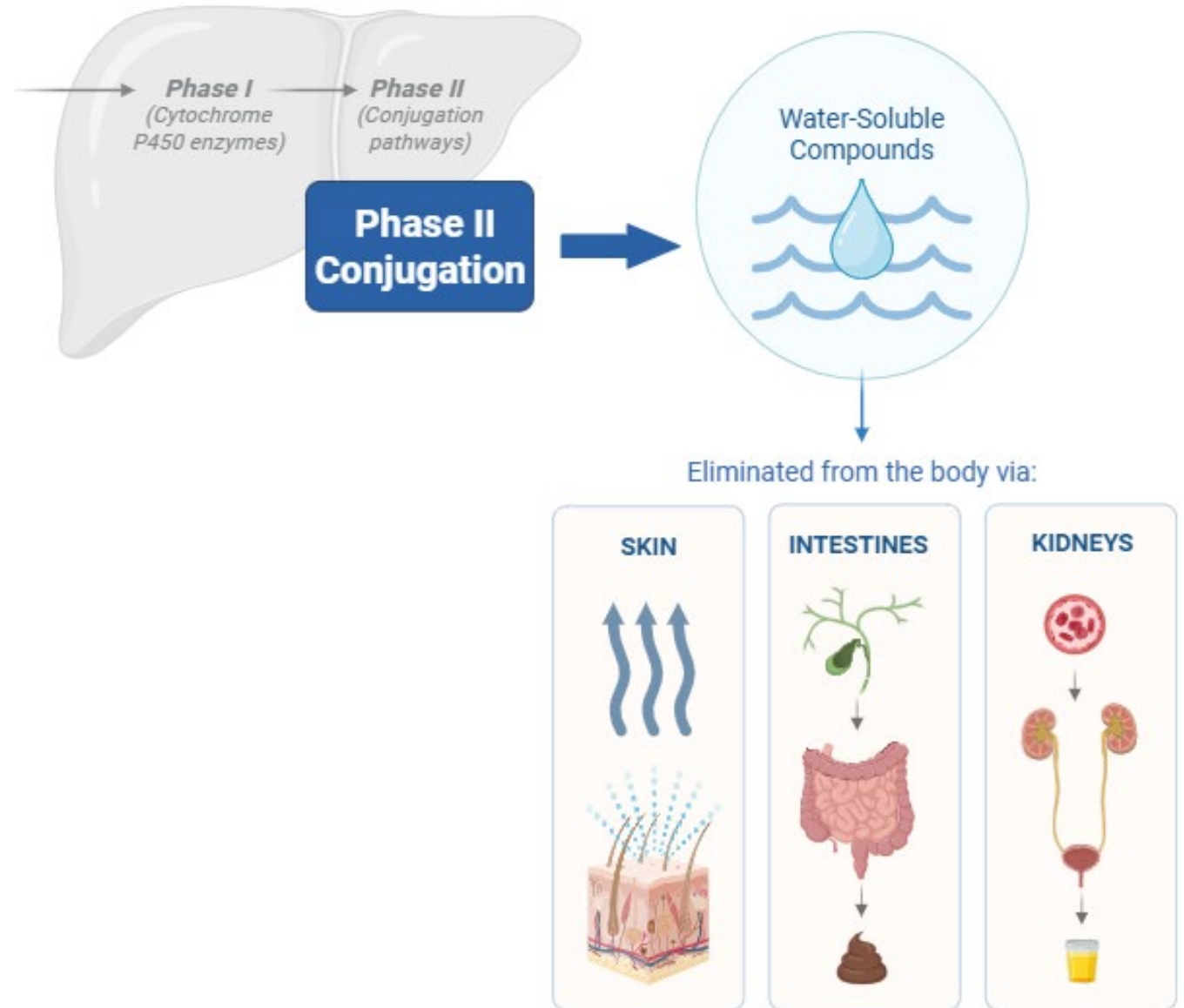
Introduce or expose a functional group on the parent compound, making it more polar.

May activate some compounds and render others *more toxic* (e.g., pro-drugs & pro-carcinogens).

Result in creation of reactive oxygen species and electrophiles, requiring consumption of antioxidant nutrients or activation of antioxidant enzymes to neutralize and mitigate.

- Susa ST, Hussain A, Preuss CV. Drug Metabolism. [Updated 2023 Aug 17]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK442023/>
- Ioannides C, Lewis DF. Cytochromes P450 in the bioactivation of chemicals. *Curr Top Med Chem*. 2004;4(16):1767-1788. doi:10.2174/1568026043387188
- Gonzalez FJ. Role of cytochromes P450 in chemical toxicity and oxidative stress: studies with CYP2E1. *Mutat Res*. 2005;569(1-2):101-110. doi:10.1016/j.mrfmmm.2004.04.021

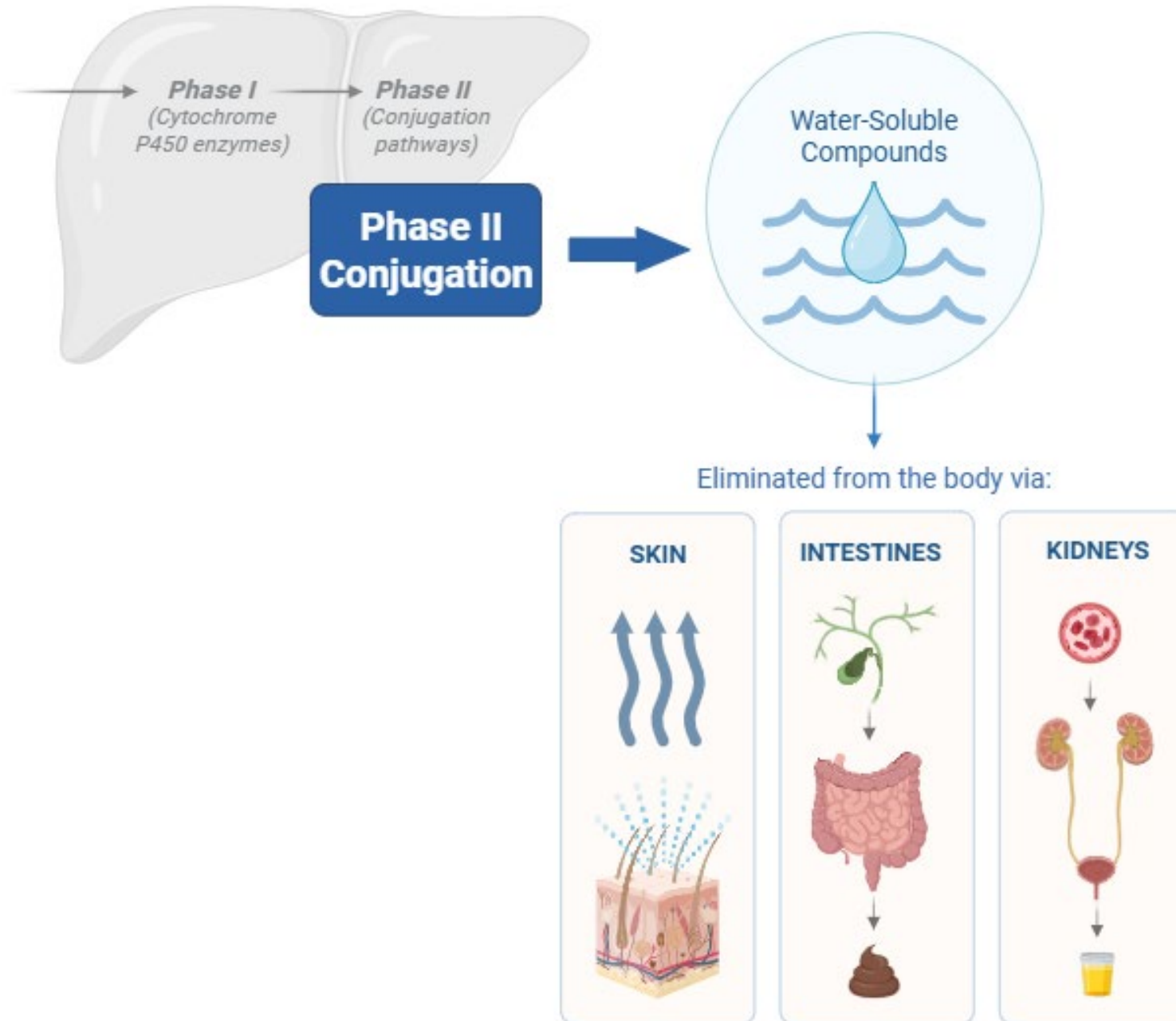
Phase II Reactions



- Covalent linkage between parent compound functional group and water-soluble moiety.
- Product inactive and excreted via bile or urine.

Long Description: Phase 2 reactions

- A diagram highlights how Phase II conjugation in the liver prepares toxins for elimination by making them water-soluble. They can then be excreted from the body.
- **Elimination Routes:** The water-soluble compounds are then excreted from the body through three primary pathways:
 - **Skin** – via sweat.
 - **Intestines** – via bile and feces.
 - **Kidneys** – via urine.
- **Summary:** After Phase II liver detoxification, water-soluble toxins are eliminated through the skin, intestines, or kidneys.



Created with BioRender.com by The Institute for Functional Medicine

Phase II Enzymes

- The conjugation process is a major consumer of cellular energy and amino acid precursors.
- Much of that energy is consumed in the production of **enzymes**, which are large globular proteins.
- Biotransformation is impaired by **UNDERNUTRITION** and may further contribute to nutrient deficiencies.

Typical Examples of Phase II Conversions

Compound	Phase I Enzyme	Conversion Product	Phase II Enzyme	Conjugate or End Product
Codeine	CYP2D6	Morphine	UGT	Glucuronide
Loratadine	CYP3A4	Desloratadine	UGT	Glucuronide
Tamoxifen	CYP2D6	4-OH-tamoxifen	UGT	Glucuronide
Acetaminophen	CYP2E1	NAPQI	GST	Glutathione (GSH)
Caffeine	CYP1A2	Paraxanthine (80-85%) Theobromine (10%)	UGT, sulfotransferase	Glucuronide, sulfate
Aflatoxin B ₁	CYP1A2, CYP3A4	Aflatoxin epoxide	GST	Glutathione (GSH)
Benzo[a]pyrene	CYP1A1/CYP1B1	Benzo[a]pyrene epoxide	GST	Glutathione (GSH)
Estradiol	CYP1A1 CYP1B1	2-OH-estradiol 4-OH-estradiol	COMT, UGT, sulfotransferase	Methoxyestradiol, glucuronide, sulfate
Alcohol	ADH (cytosol)/CYP2E1	Acetaldehyde	ALDH2	Acetate

UGT: UDP-glucuronic acid; GST: glutathione S-transferase; COMT: catechol-O-methyltransferase

See References: "Typical Examples of Phase II Conversions"

References

TYPICAL EXAMPLES OF PHASE II CONVERSIONS

- Dean L, Kane M. Codeine Therapy and CYP2D6 Genotype. 2012 Sep 20 [Updated 2025 Jan 17]. In: Pratt VM, Scott SA, Pirmohamed M, et al., editors. Medical Genetics Summaries [Internet]. Bethesda (MD): National Center for Biotechnology Information (US); 2012-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK100662/>
- Drugbank. May 30, 2025. Accessed June 3, 2025. https://www.drugbank.com/clinical/drug_search
- Tran VN, Viktorová J, Ruml T. Mycotoxins: Biotransformation and Bioavailability Assessment Using Caco-2 Cell Monolayer. *Toxins (Basel)*. 2020;12(10):628. Published 2020 Sep 30. doi:10.3390/toxins12100628
- Stoddard EG, Killinger BJ, Nag SA, et al. Benzo[a]pyrene Induction of Glutathione S-Transferases: An Activity-Based Protein Profiling Investigation. *Chem Res Toxicol*. 2019;32(6):1259-1267. doi:10.1021/acs.chemrestox.9b00069

Clinical Focus

- Effects of toxicants on physiological function in other nodes of the matrix.



Toxicity as Mediator of Dysfunction: Transport Node

Toxic metals such as lead, cadmium, mercury, arsenic, and chromium can contribute to cardiovascular disease via their ability to trigger reactive oxygen species and inflammation.

Chronic exposure to toxic metals is associated with an increased risk of hypertension, arrhythmia, and atherosclerosis.

Inhalation of fine particulate matter (PM_{2.5}) has been linked to increases in blood pressure, thrombosis, vascular injury, and acceleration of atherogenesis.

Toxicity as Mediator of Dysfunction: Defense and Repair Node

PCBs are linked to cancer
and immune suppression.

Parabens are tied to
increased inflammation
and eczema in children.

- US Environmental Protection Agency. Learn about polychlorinated biphenyls. Updated March 28, 2025. Accessed May 9, 2025. <https://www.epa.gov/pcbs/learn-about-polychlorinated-biphenyls#healtheffects>
- Peinado FM, Iribarne-Duran LM, Artacho-Cordon F. Human exposure to bisphenols, parabens, and benzophenones, and its relationship with the inflammatory response: a systematic review. *Int J Mol Sci.* 2023;24(8):7325. doi:10.3390/ijms24087325
- Arafune J, Tsujiguchi H, Hara A, et al. Increased prevalence of atopic dermatitis in children aged 0-3 years highly exposed to parabens. *Int J Environ Res Public Health.* 2021;18(21):11657. doi:10.3390/ijerph182111657

Toxicity as Mediator of Dysfunction: Energy Node

Many exogenous chemicals trigger reactive oxygen species (ROS) and oxidative stress, leading to impairment of mitochondrial function.

Examples include particulate pollution, radiation, medications, cigarette smoke, oxides of nitrogen (NO_x), and ozone (O_3).

- Jomova K, Raptova R, Alomar SY, et al. Reactive oxygen species, toxicity, oxidative stress, and antioxidants: chronic diseases and aging. *Arch Toxicol*. 2023;97(10):2499-2574. doi:10.1007/s00204-023-03562-9
- Teschke R, Xuan TD. Heavy metals, halogenated hydrocarbons, phthalates, glyphosate, cordycepin, alcohol, drugs, and herbs, assessed for liver injury and mechanistic steps. *Front Biosci (Landmark Ed)*. 2022;27(11):314. doi:10.31083/j.fbl2711314

Toxicity as Mediator of Dysfunction: Structural Integrity Node

Microplastics promote dysbiosis of the gut microbiota and intestinal permeability.

In the gut, microplastics are associated with increased oxidation and inflammation, destruction of epithelium, reduction of mucus layer, and immune cell toxicity.

- Wang X, Deng K, Zhang P, et al. Microplastic-mediated new mechanism of liver damage: from the perspective of the gut-liver axis. *Sci Total Environ.* 2024;919:170962. doi:10.1016/j.scitotenv.2024.170962
- Huang Z, Weng Y, Shen Q, Zhao Y, Jin Y. Microplastic: a potential threat to human and animal health by interfering with the intestinal barrier function and changing the intestinal microenvironment. *Sci Total Environ.* 2021;785:147365. doi:10.1016/j.scitotenv.2021.147365

Toxicity as Mediator of Dysfunction: Communication Node

Thyroid hormone synthesis, transport, metabolism, and activity may be impaired by exposure to:

Endocrine
disrupting
chemicals
(drinking water)

Air pollution

Pesticides and
herbicides

Flame
retardants

Cleaning agents

Personal care
products

Food additives
and packaging

Coatings and
solvents

Medical
products and
equipment



Long Description: toxicity as mediator of dysfunction: communication node

Thyroid hormone synthesis, transport, metabolism, and activity may be impaired by exposure to:

- Endocrine disrupting chemicals
- Air pollution
- Pesticides and herbicides
- Flame retardants
- Cleaning agents
- Personal care products
- Food additives and packaging
- Coatings and solvents
- Medical products and equipment

Toxicity as Mediator of Dysfunction: Assimilation Node



Although the liver is recognized as the main organ for biotransformation, recent studies indicate that the gut microbiota are the first to metabolize and modify the chemical structure of various ingested xenobiotics.



Diet, drugs, alcohol, and other environmental toxins can disrupt the composition of the gut microbiome and its function, leading to dysbiosis, increased gut permeability, and changes in signaling pathways.



Arsenic, cadmium, and lead (toxic metals) can alter the gut microbiome and cause inflammation through metabolite production and different signaling pathways.



Persistent organic pollutants (POPs) can alter gut microbiome via their ability to activate key signaling pathways within the intestine.

- Pant A, Maiti TK, Mahajan D, Das B. Human gut microbiota and drug metabolism. *Microb Ecol.* 2023;86(1):97-111. doi:10.1007/s00248-022-02081-x
- Teffera M, Veith AC, Ronnekleiv-Kelly S, et al. Diverse mechanisms by which chemical pollutant exposure alters gut microbiota metabolism and inflammation. *Environ Int.* 2024;190:108805. doi:10.1016/j.envint.2024.108805

Toxicants as Mediators of Dysfunction: Mental, Emotional, and Spiritual Factors

- Studies in children have linked developmental exposure to environmental chemicals with increased risk of mood, anxiety, and behavior problems.
- Lead (toxic metal) and bisphenols (endocrine disrupter in plastic) are associated with mental health illness.



Toxicants as Mediators of Dysfunction: Mental, Emotional, and Spiritual Factors

Mechanisms by which toxic metals and endocrine disruptors increase susceptibility to mental health disorders:

Alteration of
neurotransmitter systems

Disruption of blood-brain
barrier

Modulation of gene
expression (depression)

Dysregulation of the
hypothalamus-pituitary-
adrenal axis

Reduction of neuronal
plasticity

Induction of oxidative
stress and
neuroinflammation

Clinical Focus

- Effects of dysfunction on other nodes of the matrix that contribute to an elevated total toxic load.



Transport Node:

Obesity as a Mediator of Impaired Detoxification

Obesity is linked to decreased levels of cytochrome P450 (CYP) enzymes and impaired biotransformation of toxicants.



Communication Node:

PCOS and MASLD: Mechanisms That Increase Total Toxin Load

CYP450 liver enzymes are involved in steroid hormone detoxification and metabolism.

MASLD, commonly associated with PCOS, leads to impaired activity of CYP450, contributing to impaired steroid hormone metabolism.

This impairment of CYP450 activity can potentially lead to the elevated androgen levels that are a hallmark of PCOS.

Structural Integrity Node:

Intestinal Permeability: Mechanisms That Increase Toxic Load

In normal conditions, a healthy intestinal barrier prevents toxic substances from entering the bloodstream.

Increased intestinal permeability from any cause results in an increased toxic load via this mechanism.

Defense and Repair Node:

Inflammation as Mediator of Increased Toxicant Effects

Infection and inflammation lead to a decrease in the expression and activity of cytochrome P450 enzymes, which are crucial for biotransformation of xenobiotics.

Pro-inflammatory cytokines, which are released during immune responses, are major mediators of the suppression of CYP enzymes necessary to clear toxins.

Assimilation Node:

Dysbiosis as a Mediator of Increased Toxic Load

Gut microbiome metabolism can modulate the toxicity of certain toxins.

Xenobiotics excreted by liver into bile and then into intestines are metabolized by GI microbiota.

The microbiota deconjugates and processes xenobiotic metabolites, resulting in reabsorbable, lower molecular weight, nonpolar molecules.

Dysbiotic species increase the likelihood of deconjugation and formation of toxic metabolites.

- Claus SP, Guillou H, Ellero-Simatos S. The gut microbiota: a major player in the toxicity of environmental pollutants? [published correction appears in *NPJ Biofilms Microbiomes*. 2017;3:17001. doi:10.1038/npjbiofilms.2017.1]. *NPJ Biofilms Microbiomes*. 2016;2:16003. doi:10.1038/npjbiofilms.2016.3
- Koontz JM, Dancy BCR, Horton CL, Stallings JD, DiVito VT, Lewis JA. The role of the human microbiome in chemical toxicity. *Int J Toxicol*. 2019;38(4):251-264. doi:10.1177/1091581819849833

Clinical Focus

- Therapeutic principles for addressing elevated total toxic load: reduce exposures.



Susceptibility:

Factors That Need to Be Addressed

- Increased or ongoing exposure (toxic threshold exceeded)
- Nutrient deficiencies (B vitamins, antioxidants, minerals, etc.)
- Highly refined/processed food intake, inadequate protein
- Stress, emotional trauma
- Intestinal dysbiosis
- **Genetic variants in phase I/II (polymorphisms/SNPs)**



References

WHY SOME PEOPLE RETAIN OR ARE MORE SENSITIVE TO TOXINS

Highly Processed Diet:

- Nogacka AM, Gómez-Martín M, Suárez A, González-Bernardo O, de Los Reyes-Gavilán CG, González S. Xenobiotics formed during food processing: their relation with the intestinal microbiota and colorectal cancer. *Int J Mol Sci*. 2019;20(8):2051. doi:10.3390/ijms20082051

Insufficient Protein:

- Hou Y, Yin Y, Wu G. Dietary essentiality of "nutritionally non-essential amino acids" for animals and humans. *Exp Biol Med (Maywood)*. 2015;240(8):997-1007. doi:10.1177/1535370215587913

Nutrition:

- Hoffman JB, Hennig B. Protective influence of healthful nutrition on mechanisms of environmental pollutant toxicity and disease risks. *Ann N Y Acad Sci*. 2017;1398(1):99-107. doi:10.1111/nyas.13365

Gut Dysbiosis:

- Rosenfeld C. Gut dysbiosis in animals due to environmental chemical exposures. *Front Cell Infect Microbiol*. 2017;7:396. doi:10.3389/fcimb.2017.00396
- Claus S, Guillou H, Ellero-Simatos S. The gut microbiota: a major player in the toxicity of environmental pollutants? *NPJ Biofilms Microbiomes*. 2016;2:16003. doi:10.1038/npjbiofilms.2016.3

References

WHY SOME PEOPLE RETAIN OR ARE MORE SENSITIVE TO TOXINS

Genetic Variations/SNPs:

- Marin JJG, Serrano MA, Monte MJ, et al. Role of genetic variations in the hepatic handling of drugs. *Int J Mol Sci*. 2020;21(8):2884. doi:10.3390/ijms21082884
- Bukowski K, Woźniak K. Polymorphism of genes encoding proteins of DNA repair vs. occupational and environmental exposure to lead, arsenic and pesticides. *Med Pr*. 2018;69(2):225-235. doi:10.13075/mp.5893.00595

Stress/Emotional Trauma:

- Deng Q, Deng L, Lu C, Li Y, Norbäck D. Parental stress and air pollution increase childhood asthma in China. *Environ Res*. 2018;165:23-31. doi:10.1016/j.envres.2018.04.003

Nutrient Deficiencies:

- Miller CN, Rayalam S. The role of micronutrients in the response to ambient air pollutants: potential mechanisms and suggestions for research design. *J Toxicol Environ Health B Crit Rev*. 2017;20(1):38-53. doi:10.1080/10937404.2016.1261746
- Baccarelli A, Cassano PA, Litonjua A, et al. Cardiac autonomic dysfunction: effects from particulate air pollution and protection by dietary methyl nutrients and metabolic polymorphisms. *Circulation*. 2008;117(14):1802-1809. doi:10.1161/CIRCULATIONAHA.107.726067

Ongoing Exposures and Increased Risk of Mortality and Chronic Disease:

- Bauleo L, Bucci S, Antonucci C, et al. Long-term exposure to air pollutants from multiple sources and mortality in an industrial area: a cohort study. *Occup Environ Med*. 2019;76(1):48-57. doi:10.1136/oemed-2018-105059

Total Toxic Load Is Determined by:

Total Exposure
(to Toxic
Substances)



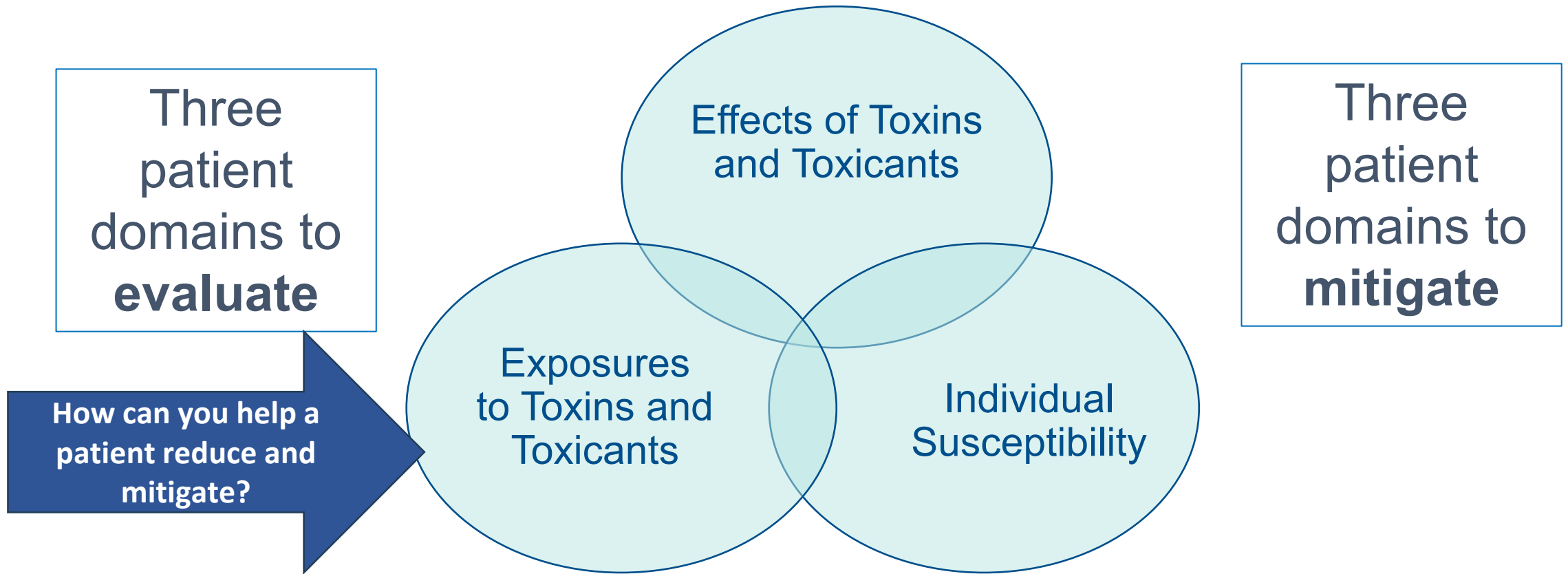
Ability to
Mitigate,
Biotransform, or
Metabolize



Clearance or
Elimination

The “Toxic Triad”

Susceptibility – Exposure – Effect



AVOID: Minimizing Toxic Exposures

AVOID

MITIGATE

CLEAR

AVOID: Minimizing Toxic Exposures

FOOD



- Fresh is best – minimize processed foods, packaged foods, and fast foods.
- Eat organically grown fruits and vegetables when possible.
- If omnivorous, emphasize organic meat, poultry, and dairy products.
- If pescatarian, eat low-mercury fish: salmon, sardines, anchovies, trout, herring, cod.
- Try to identify source of shellfish (avoid oysters downstream from wastewater).

WATER



- Avoid plastic water bottles, especially single-use plastic.
- Emphasize fresh spring water when possible; otherwise drink purified water in glass, ceramic-lined, or stainless-steel bottles.
- Eat fresh foods instead of foods that are processed, canned, or wrapped in plastic.

AVOID: Minimize Toxic Exposure

Skin

- Use “natural” cosmetics, shampoos, nail polish, and fragrances.
- Avoid contact with toxic solvents and cleaning agents: use vinegar, baking soda, Bon Ami (feldspar)
- Avoid contact with synthetic pesticides/herbicides

Mouth

- Use dentists who are aware of potential toxic effects of amalgams and plasticizers

Liver

- Low to moderate alcohol consumption; avoid daily use
- Minimize use of acetaminophen (especially after drinking alcohol)

Lungs

- Use nontoxic building materials and carpets when remodeling or building a new home

In Your Toolkit



Reducing Exposures to Harmful Chemicals

Daily exposure to potentially harmful chemicals is virtually unavoidable. They're in the food you eat, the water you drink, and even in the personal care products you use every day. It is easy to feel overwhelmed by the sheer number of exposures. The good news is that small, intentional diet and lifestyle changes can make a big difference. This guide is here to help you take practical steps to reduce your harmful chemical exposure and take charge of your health.

Chemicals Are Linked to Disease Risk

Seven categories of chemicals are especially common and associated with harmful health effects:

Type of Chemical	Definitions and Examples	Potential Harmful Health Effects
Toxic metals	Harmful metals that build up in your body over time Examples include arsenic, cadmium, lead, and mercury.	Exposure to high levels of toxic metals can be harmful to your body.¹ Low to moderate exposure is linked to many health issues, including cardiovascular disease,² chronic kidney disease,³ cancer,⁴ changes in thyroid hormone levels,⁵ and thyroid cancer.⁶
Plastics and plasticizers	Plastics are used in many consumer products, like water bottles and food containers. Examples include polycarbonate and polystyrene. Plasticizers are mixed with plastics to make them flexible or durable. They are also found in personal care products like makeup and lotion.⁷ Examples include bisphenol A (BPA) and phthalates. Microplastics are tiny pieces of plastic debris that result from plastic breakdown.	BPA and phthalates are connected to obesity,^{8,9} metabolic syndrome,¹⁰ and diabetes.¹¹ Polystyrene is linked to liver and kidney cell inflammation.¹² Microplastics are linked to impaired fetal growth,¹³ harmful effects on sperm health,¹⁴ and cardiovascular disease.¹⁵
Pesticides and herbicides	Chemicals used in agriculture and lawn care Pesticides kill insects, while herbicides kill weeds. Examples include organochlorine (OC) and organophosphate (OP) pesticides and glyphosate, a common herbicide.	Pesticide exposure is connected to Parkinson's disease,¹⁶ cardiovascular disease,¹⁷ colon cancer,¹⁸ harmful effects on sperm health,¹⁹ asthma, wheezing, and diabetes.²⁰ Herbicide exposure is linked to cancer²¹ and reduced fertility.²²

Environmental Health: Part 5

Clinical Approach to Total Toxic Load

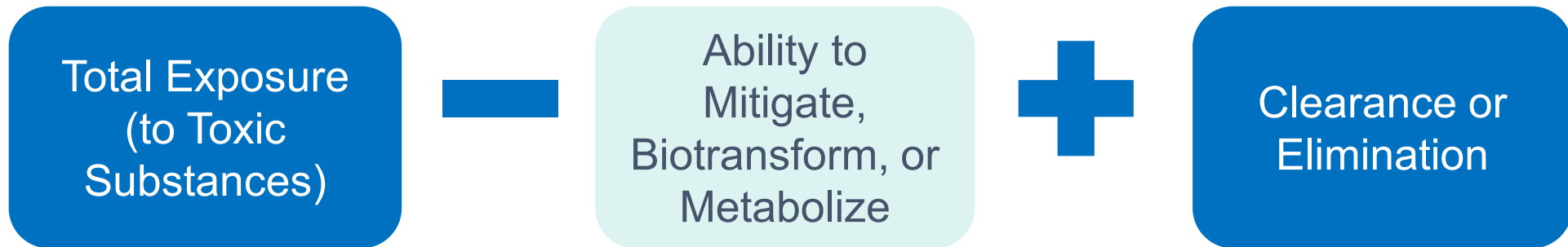
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Therapeutic principles for addressing elevated total toxic load: address undernourishment to reduce susceptibility.

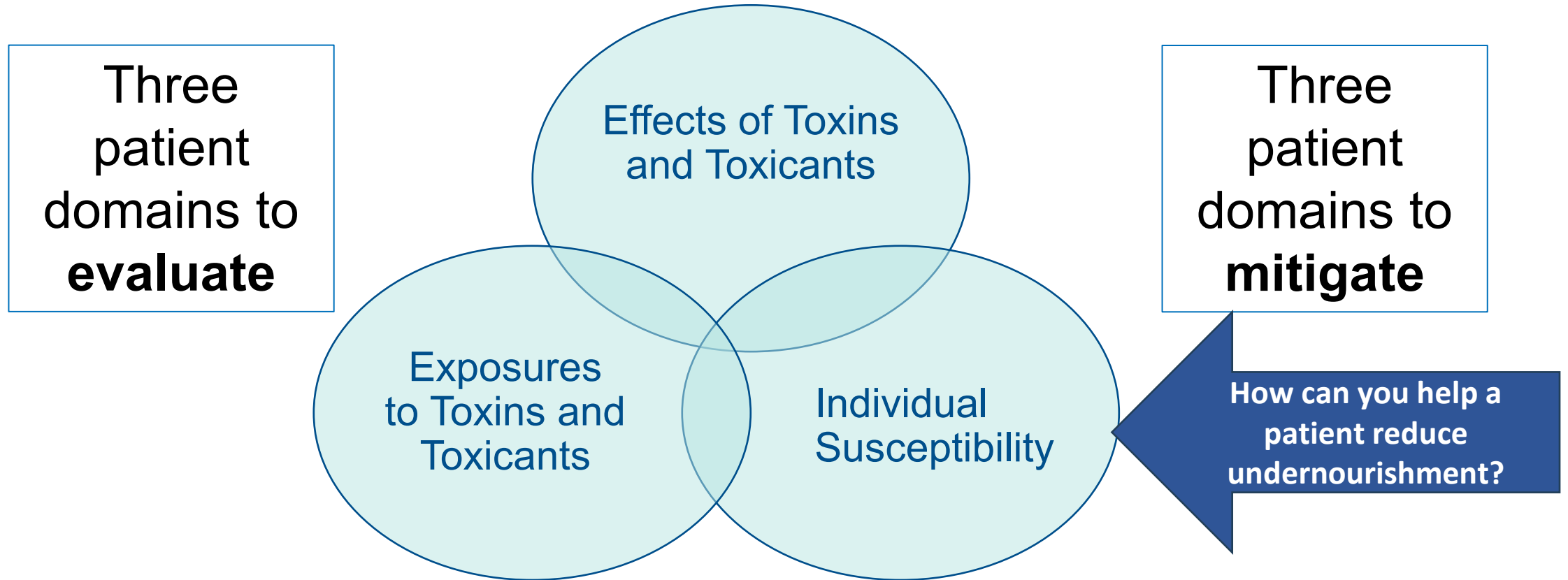


Total Toxic Load Is Determined by:



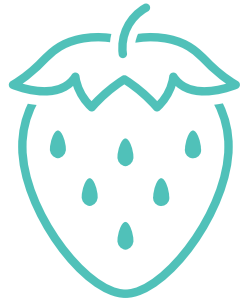
The “Toxic Triad”

Susceptibility – Exposure – Effect



Nutrition Status:

Modulator of Susceptibility to Environmental Pollutants



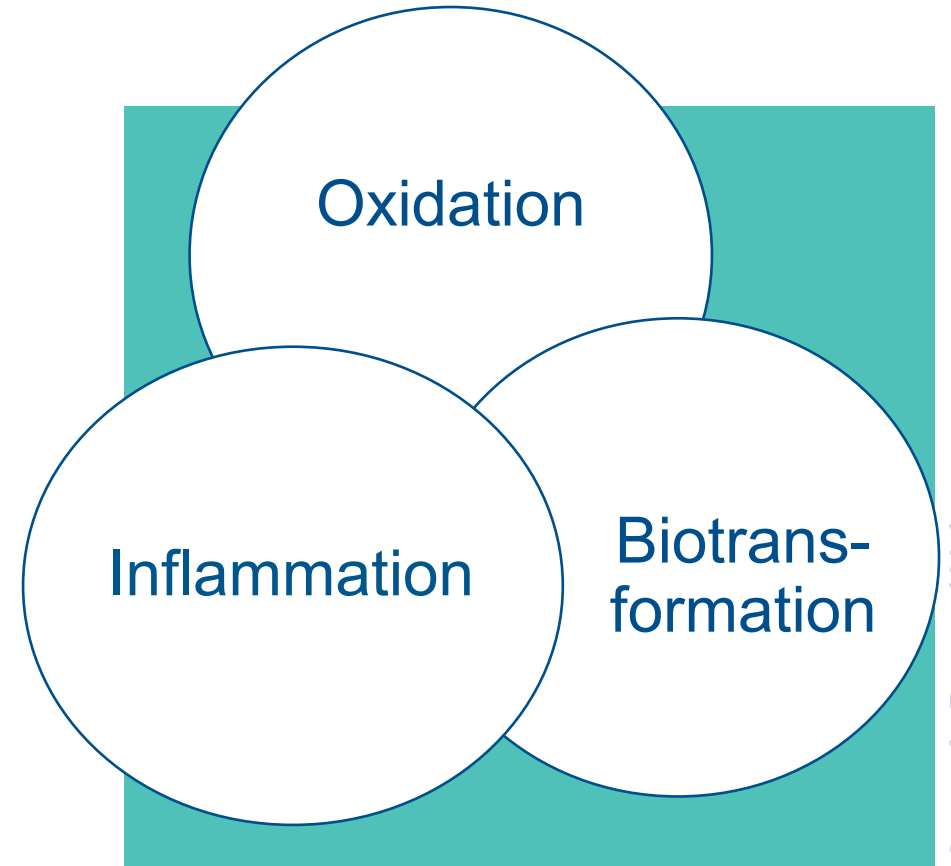
A convincing body of research indicates that **nutrition is a modulator of susceptibility to environmental insults**; thus, it is essential to consider nutritional status as a vital component of human risk assessment.



Dietary intake may serve as either an agonist or an antagonist of susceptibility to the health impacts associated with exposure to environmental pollutants.

Reducing Susceptibility to Toxic Substances: The “Triple Aim” of Functional Nutrition

- Enhance antioxidant capacity
- Downregulate inflammation
- Balance phase I & II biotransformation pathways



Reducing Susceptibility to Toxic Substances: The “Triple Aim” of Functional Nutrition

- **Enhance antioxidant capacity**
- Downregulate inflammation
- Balance phase I & II biotransformation pathways



Enhance Antioxidant Capacity: Reduce Susceptibility to Effects of Toxic Substances

- Study data suggest that antioxidant nutrients, such as vitamin E and dietary flavonoids, as well as a high ratio of omega-3 to omega-6 fatty acids, can protect against endothelial cell damage mediated by persistent organic pollutants.
- **Diets rich in antioxidants and anti-inflammatory nutrients** can improve health and decrease vulnerability to additional chemical stressors.

Reducing Susceptibility to Toxic Substances: The “Triple Aim” of Functional Nutrition

- Enhance antioxidant capacity
- **Downregulate inflammation**
- Balance phase I & II biotransformation pathways



Downregulate Inflammation: Reduce Susceptibility to Effects of Toxic Substances



Anti-inflammatory nutrients, in particular **dietary long-chain omega-3 polyunsaturated fatty acids and plant-derived flavonoids**, can mitigate toxicological insults and pathologies associated with exposure to environmental pollutants.



Fish-derived omega-3 fatty acids may have a protective role in pollutant-induced inflammation from toxic metals such as cadmium.

- Chen Z, Lu Q, Wang J, et al. The function of omega-3 polyunsaturated fatty acids in response to cadmium exposure. *Front Immunol.* 2022;13:1023999. doi:10.3389/fimmu.2022.1023999
- Djuricic I, Calder PC. Beneficial outcomes of omega-6 and omega-3 polyunsaturated fatty acids on human health: an update for 2021. *Nutrients.* 2021;13(7):2421. doi:10.3390/nu13072421

Reducing Susceptibility to Toxic Substances: The “Triple Aim” of Functional Nutrition

- Enhance antioxidant capacity
- Downregulate inflammation
- **Balance phase I & II
biotransformation pathways**



Balance Phase I & II Biotransformation: Reduce Susceptibility to Effects of Toxicants

- Adequate **protein source** for enzyme production
- **Conjugant precursors** for phase II reactions
- **Enzymatic cofactors** for phase I & II
- **Transcription inducers** of phase II enzymes

Clinical Focus

- Lifestyle changes as safe and effective interventions to reduce total toxic load.



Foundational Therapeutic Approach: Mitigation & Clearance – Reduce Total Toxic Load



LIFESTYLE FIRST



FOOD FIRST



GUT FIRST

Total Toxic Load Is Determined by:

Total Exposure
(to Toxic
Substances)



Ability to
Mitigate,
Biotransform, or
Metabolize



Clearance or
Elimination

Lifestyle Factors to Reduce Toxic Load



Sleep Interventions to Reduce Toxic Load



Animal Studies: Toxin Clearance via Glymphatics

In one study, researchers injected mice with beta-amyloid, a protein associated with Alzheimer's disease, and measured how long it lasted in their brains when they were asleep or awake.

Beta-amyloid disappeared faster in mice brains when the mice were asleep, suggesting that sleep quality may mitigate the effects of toxicants in the brain.

- Chong PLH, Garic D, Shen MD, Lundgaard I, Schwichtenberg AJ. Sleep, cerebrospinal fluid, and the glymphatic system: a systematic review. *Sleep Med Rev.* 2022;61:101572. doi:10.1016/j.smrv.2021.101572
- Miao A, Luo T, Hsieh B, et al. Brain clearance is reduced during sleep and anesthesia [published correction appears in *Nat Neurosci.* 2024;27(7):1425. doi:10.1038/s41593-024-01698-0]. *Nat Neurosci.* 2024;27(6):1046-1050. doi:10.1038/s41593-024-01638-y
- Xie L, Kang H, Xu Q, et al. Sleep drives metabolite clearance from the adult brain. *Science.* 2013;342(6156):373-377. doi:10.1126/science.1241224

Exercise Interventions to Reduce Toxic Load



In a study of 200 participants aged 17-19 who engaged in different forms of physical activity for three months (3x/week, 90 min sessions), a decrease in levels of the following toxins were observed:

- Dimethoate (organophosphate insecticide)
- BaP (polycyclic aromatic hydrocarbon)
- BPA (type of plastic)



Nutrition Interventions to Reduce Toxin Load



Evidence suggests that poor diet and malnutrition may increase the risk for oxidative stress and chronic diseases.



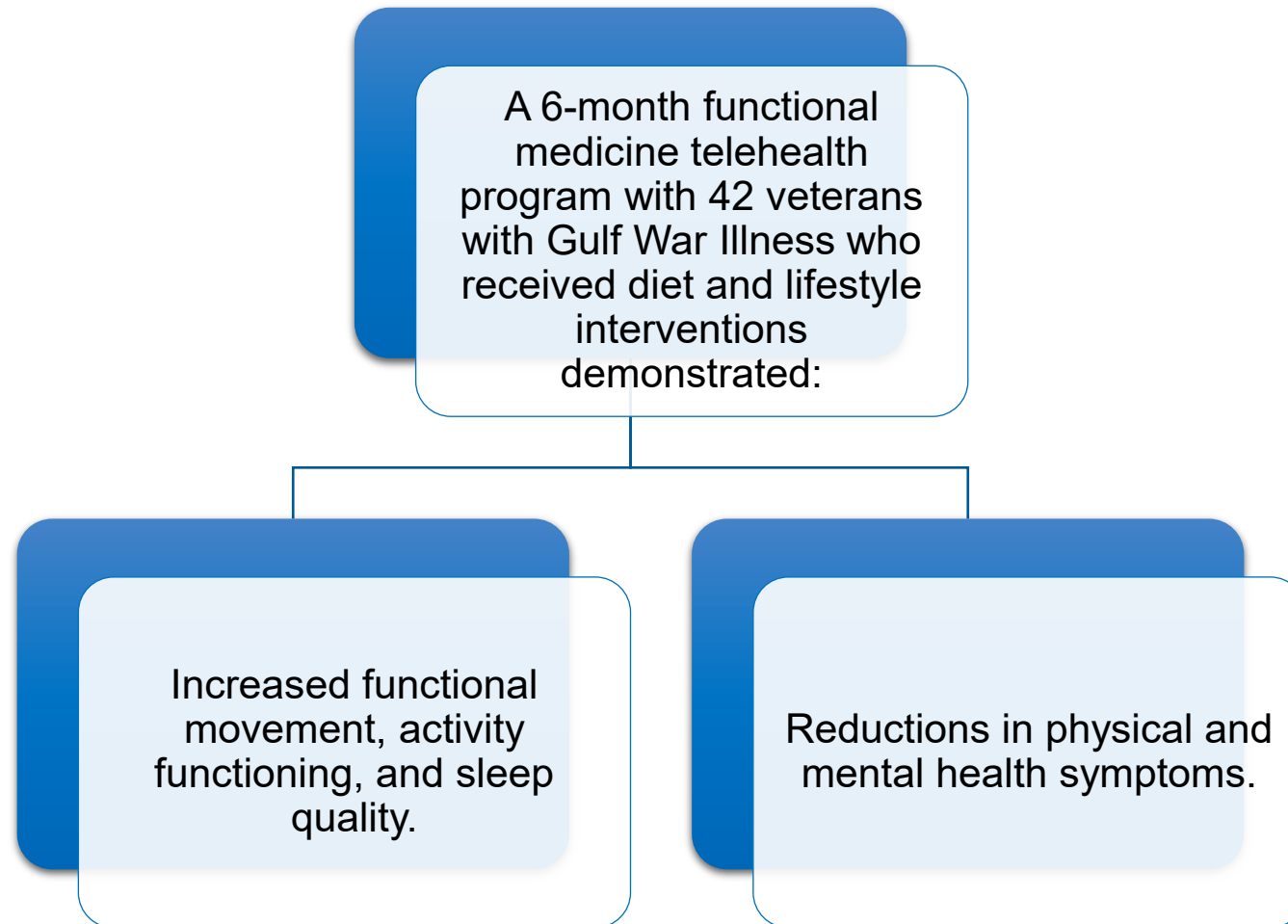
Nutrients such as carotenoids, vitamins D, E, and C, curcumin, choline, and omega-3 fatty acids may play a protective role against pollution-induced oxidative stress that can trigger asthma, COPD, and lung cancer.



The Mediterranean diet, which is rich in flavonoids, can help mitigate health risks from exposure to environmental pollutants.

- Whyand T, Hurst JR, Beckles M, Caplin ME. Pollution and respiratory disease: can diet or supplements help? A review. *Respir Res.* 2018;19(1):79. doi:10.1186/s12931-018-0785-0
- Montano L, Maugeri A, Volpe MG, et al. Mediterranean diet as a shield against male infertility and cancer risk induced by environmental pollutants: a focus on flavonoids. *Int J Mol Sci.* 2022;23(3):1568. doi:10.3390/ijms23031568

Functional Medicine Diet and Lifestyle Interventions to Support all MLFs



Clinical Focus

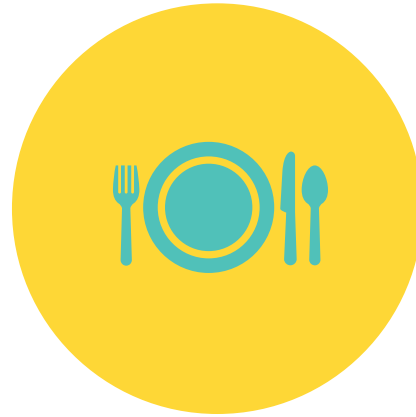
- Therapeutic foods and food plans for addressing elevated total toxic load.



Foundational Therapeutic Approach: Mitigation & Clearance – Reduce Total Toxic Load



LIFESTYLE FIRST

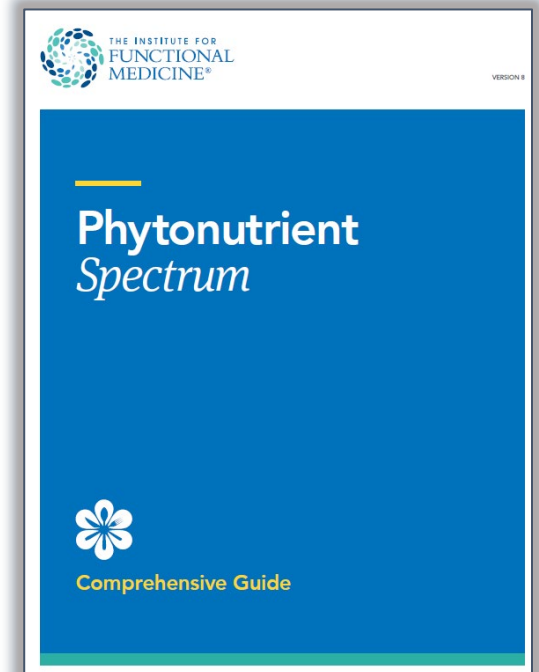
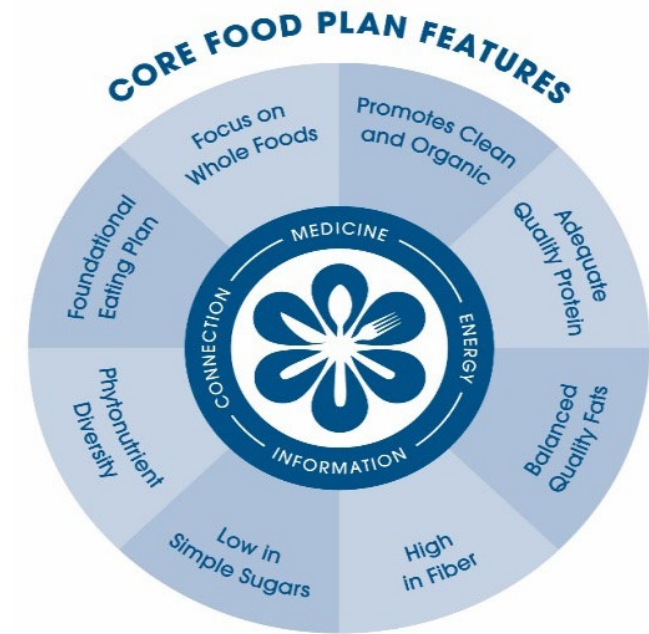
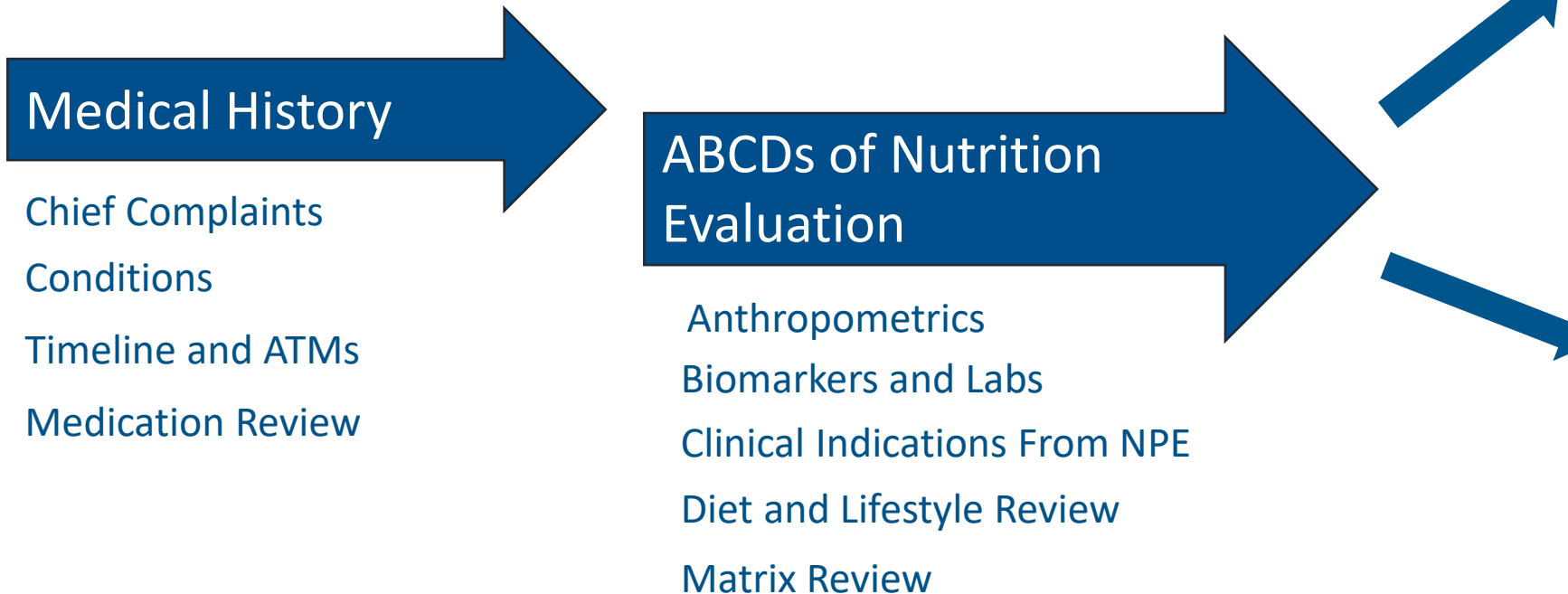


FOOD FIRST



GUT FIRST

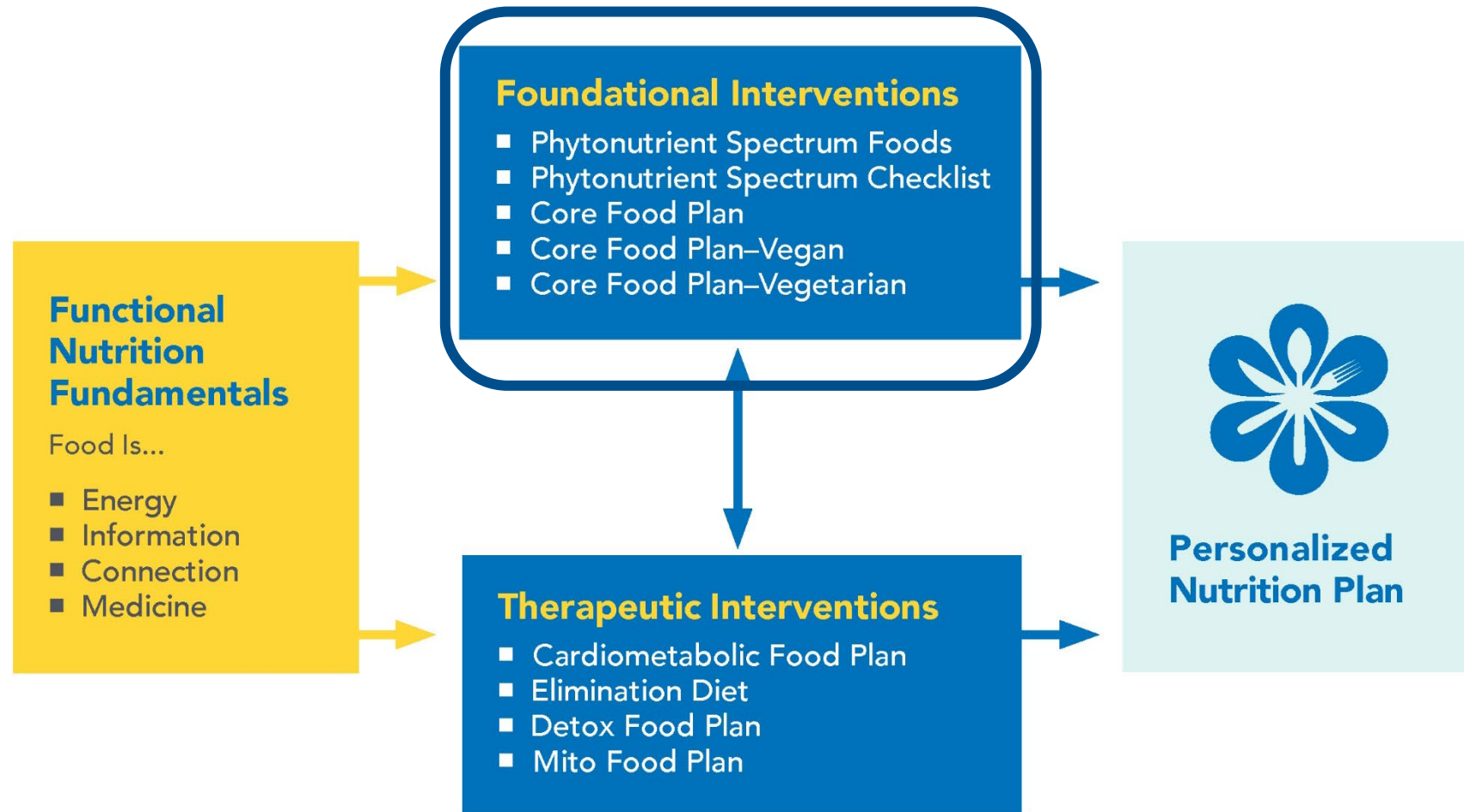
The Clinical Path to the Foundational Food Plans



Long Description: The clinical path to the foundational food plans

- A graphic outlines the clinical steps leading to the implementation of foundational food plans in functional medicine.
- **Step 1 – Medical History:** The process begins with collecting medical history, including:
 - Chief complaints
 - Existing conditions
 - Personal timeline and antecedents/triggers/mediators (ATMs)
 - Medication review
- **Step 2 – A B C Ds of Nutrition Evaluation:** This includes:
 - Anthropometrics (measurements of body size/shape)
 - Biomarkers and lab results
 - Clinical indications from the Nutrition Physical Exam (NPE)
 - Diet and lifestyle review
 - Functional Medicine Matrix review
- Based on the above steps of Medical History and A B C Ds of Nutrition, appropriate foundational food plans are selected. These include the Core Food Plan and the Phytonutrient Spectrum.

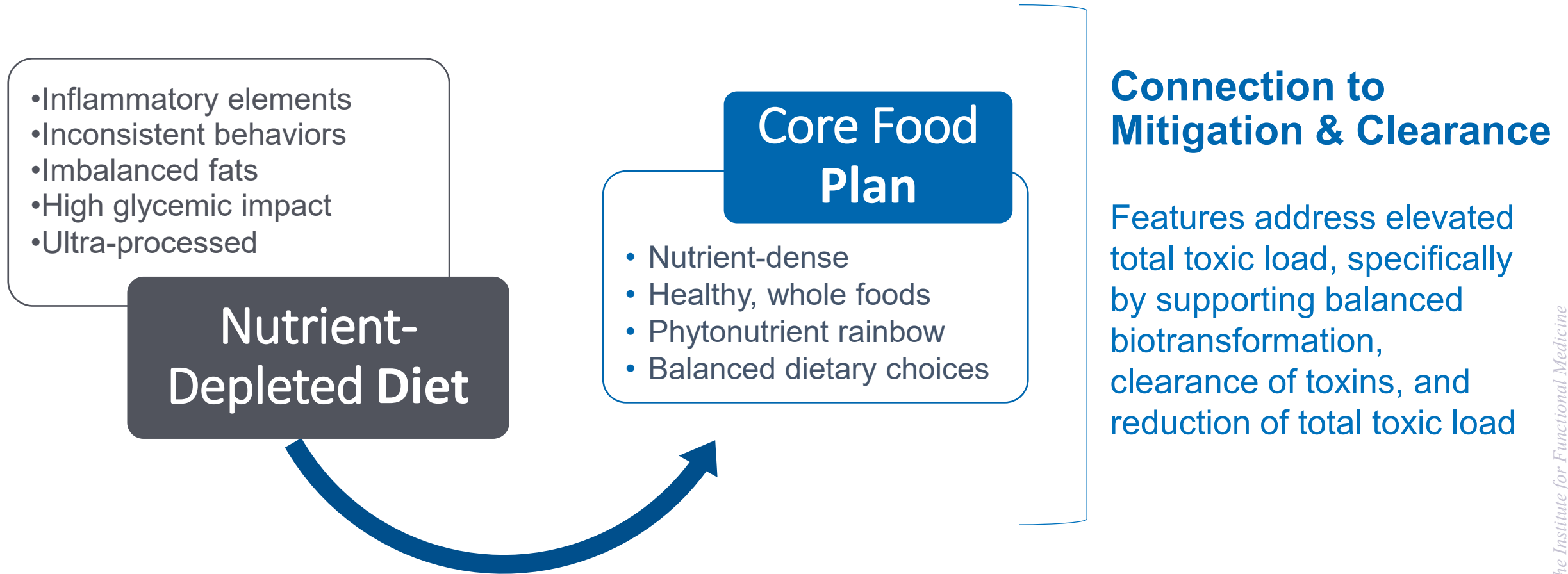
Nutrition Interventions



Long Description: Nutrition Interventions

- A diagram outlines the framework for developing a personalized nutrition plan in functional medicine through foundational and therapeutic interventions. On this slide, Foundational Interventions are emphasized
- **Left Column – Functional Nutrition Fundamentals**
States that food serves four key roles:
 - Energy
 - Information
 - Connection
 - Medicine
- These fundamentals guide both foundational and therapeutic nutrition intervention strategies.
- **Center – Two Types of Interventions in boxes**
 - **Foundational Interventions** (top box, the one emphasized on this slide):
 - Phytonutrient Spectrum Foods
 - Phytonutrient Spectrum Checklist
 - Core Food Plan
 - Core Food Plan – Vegan
 - Core Food Plan – Vegetarian
 - **Therapeutic Interventions** (bottom box):
 - Cardiometabolic Food Plan
 - Elimination Diet
 - Detox Food Plan
 - Mito Food Plan
- **Right Column – Personalized Nutrition Plan**
The foundational and therapeutic interventions together lead to the creation of a **Personalized Nutrition Plan** tailored to the individual.
- **Summary:**
Personalized nutrition in functional medicine begins with core nutrition principles and applies foundational and therapeutic food plans to support individual health needs.

Recommended “Core” First Step for Most Patients: A more nutrient-dense diet

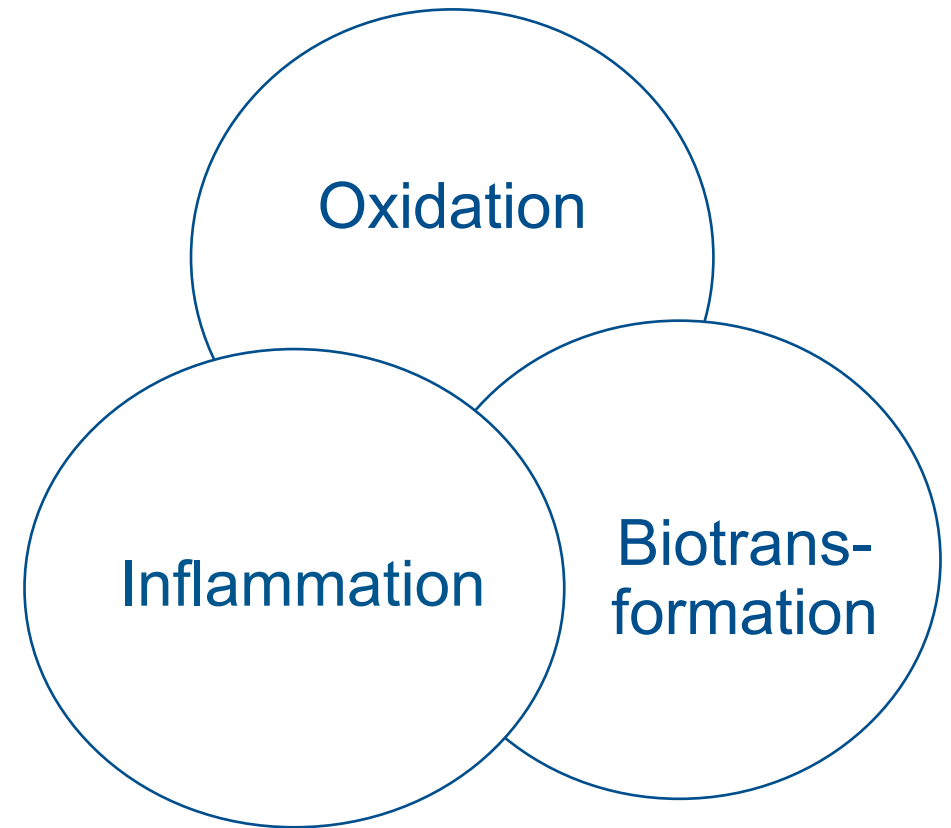
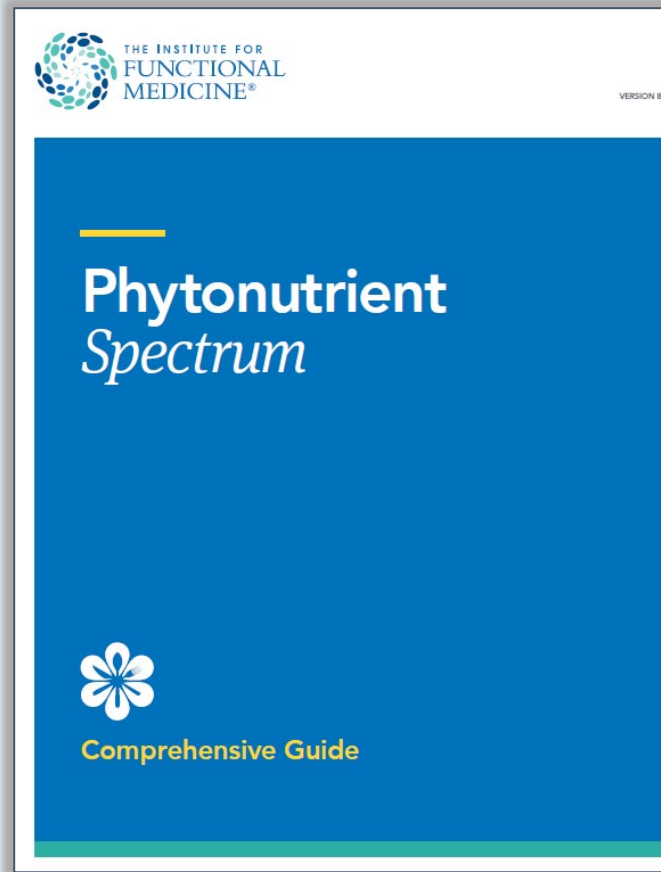


Long Description: Recommended Core first step for most patients. A more nutrient dense diet

- A graphic explains the importance of shifting from a nutrient-depleted diet to a nutrient-dense diet as a core first step for most patients.
- On the left, a "Nutrient-Depleted Diet" is described as containing inflammatory elements, inconsistent behaviors, imbalanced fats, high glycemic impact, and ultra-processed foods. These lead one toward the "Core Food Plan," which includes nutrient-dense, healthy whole foods, a phytonutrient rainbow (variety of colorful plant foods), and balanced dietary choices.
- To the right, the "Connection to Mitigation & Clearance" is emphasized. It explains that a nutrient-dense diet as found in the Core Food Plan helps address elevated total toxic load by supporting balanced biotransformation, detoxification, and overall reduction of toxic burden in the body.
- The overall message is that improving diet quality is a foundational step to support detoxification and health.

Phytonutrient Spectrum

Targets the “Triple Aim” to Safely Reduce Total Toxin Load



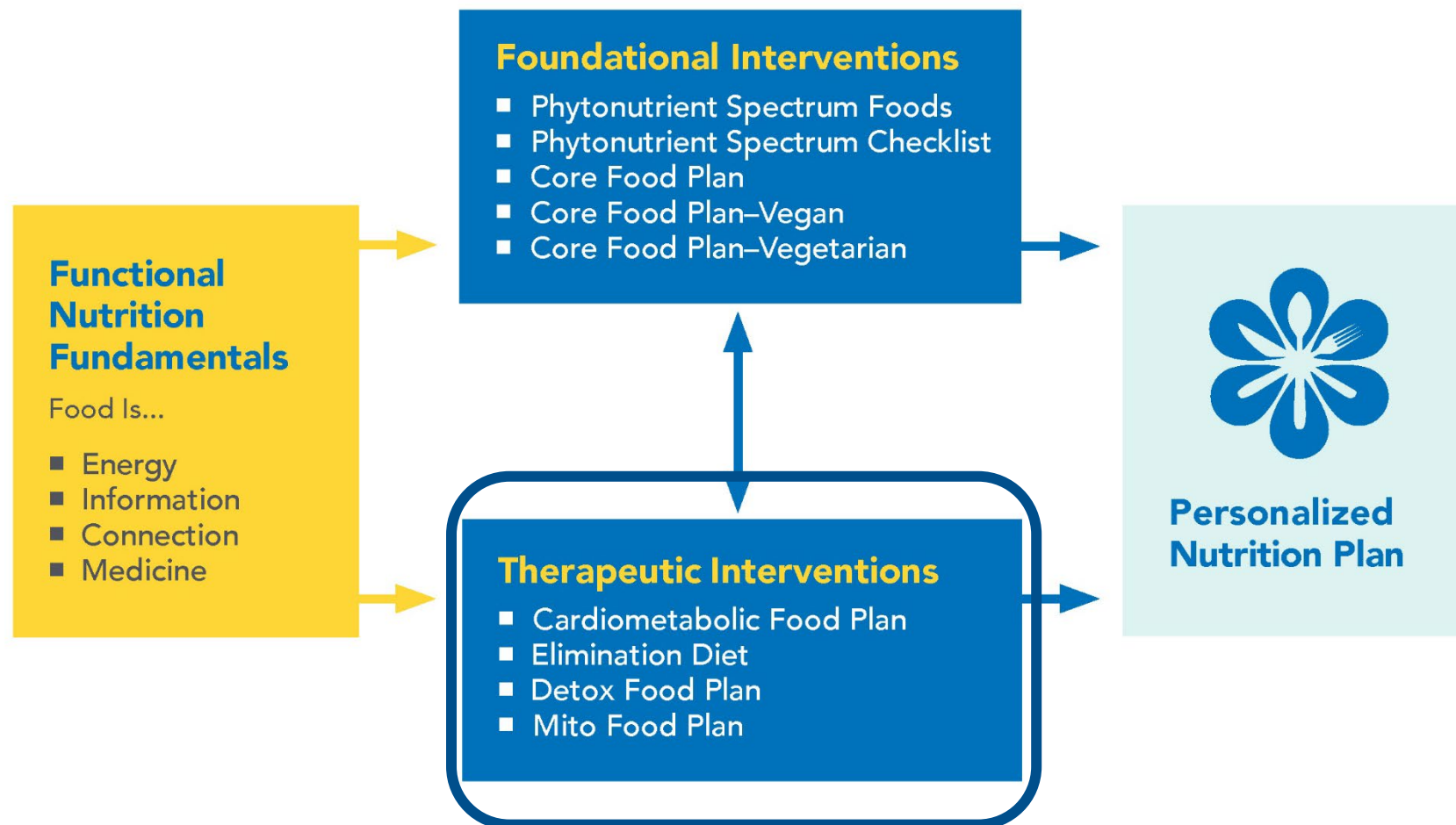
The Problem, the Solution, and the Goal

- Low-grade inflammation and insulin resistance play key roles in the pathophysiology of cardiovascular and metabolic diseases.
- A low-glycemic, anti-inflammatory, and phytonutrient-rich diet can help to stabilize blood sugar levels, decrease chronic inflammation, and reduce insulin secretion.



As the patient's blood sugar stabilizes, both cardiovascular and metabolic disease are mitigated.

Nutrition Interventions



Long Description: Nutrition Interventions

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The foundational and therapeutic interventions together lead to the creation of a **Personalized Nutrition Plan** tailored to the individual.
- **Summary:**
Personalized nutrition in functional medicine begins with core nutrition principles and applies foundational and therapeutic food plans to support individual health needs.

What Is the Detox Food Plan?

A clinically directed metabolic detoxification protocol used to:

- Facilitate the pathways involved in the processing and excretion of toxins.
- Improve symptoms of pain and fatigue.
- Enhance cognitive function and moods.
- Improve sleep quality.
- Optimize the metabolic detoxification experience and lower the toxic burden.



Detox Food Plan Features

1. Supports liver function
2. Encourages organic
3. Reduces toxic burden
4. Provides targeted antioxidants
5. Supports nutrient-dependent pathways
6. Encourages healthy elimination of toxins
7. Balances hormone metabolism
8. Reduces food triggers



Core & Detox Food Plan Features

Learn more about
the Detox Food Plan
in the Environmental
Health APM

Core Food Plan	Detox Food Plan
Focus on Whole Foods & Phytonutrient Diversity	Supports Liver Function, Reduces Toxic Burden, Supports Nutrient-Dependent Pathways, Provides Targeted Antioxidants, Encourages Healthy Elimination of Toxins
Encourages Organic	Encourages Organic, Reduces Toxic Burden
Adequate Quality Protein	Supports Liver Function, Encourages Healthy Elimination of Toxins
Balanced Quality Fats	Reduces Toxic Burden, Supports Liver Function, Encourages Healthy Elimination of Toxins
High Fiber	Reduces Toxic Burden, Encourages Healthy Elimination of Toxins
Low in Simple Sugars	Reduces Toxic Burden
Dairy-Free, Gluten-Free (optional)	Reduces Food Triggers

Superfoods:

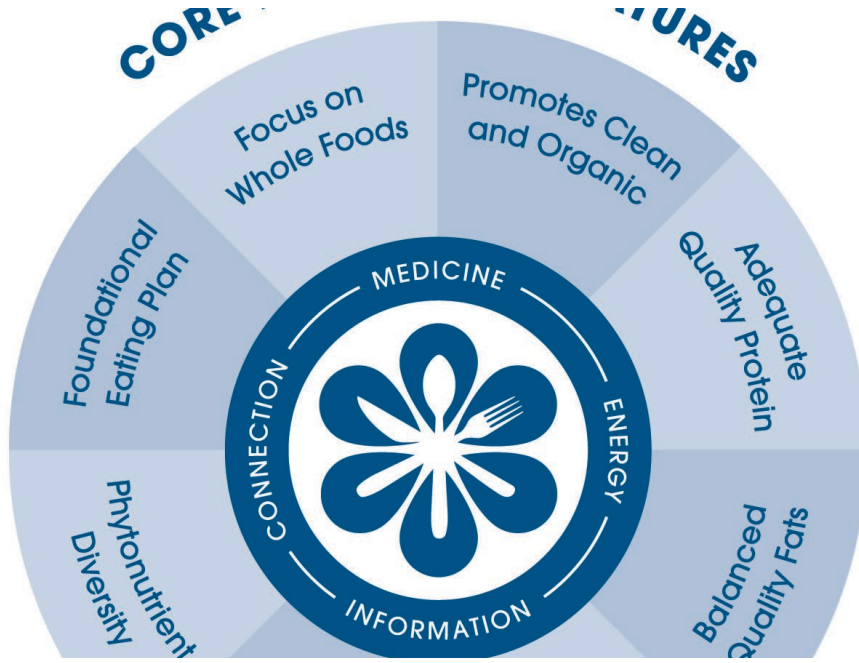
Effective Foundational Interventions for Most Patients

- Fermented foods
- Fiber: oats, flax meal, chia seeds
- Chlorella (green algae)
- Brassica vegetables: broccoli, watercress, cabbage, kale
- Green tea
- Spices: garlic, rosemary, curry
- Berries and pomegranate
- Omega-3 foods (“SMASH” fish et al)

Foundational Botanicals From Food Sources

Effective Foundational Interventions for Most Patients

Phytonutrient	Sources	Detox Effects
Curcumin (diferuloylmethane)	Turmeric	• Upregulation of phase II • Cholagogue & choleretic • Antioxidant & anti-inflammatory
Glucosinolates: • Sulforaphane • Phenylethylisothiocyanate • Diindolylmethane	Cruciferous vegetables • Cabbage • Broccoli • Brussels sprouts • Watercress	• Induces phase II detox enzymes • Induces antioxidant enzymes
Ellagic Acid	Walnuts, pomegranate, raspberries, strawberries	• Hepatoprotectant • Binds some toxins • Inhibits CYP1A1 of carcinogens • Induces phase II
Catechins	Green Tea Cocoa	• Bifunctional modulator of phase I and II enzymes • Antioxidant • Binds toxins
Organosulphates • Gamma-glutamylcysteines • Cysteine sulfoxides	Alliums • Garlic • Onion	• Increases glutathione • Induces phase II



Clinical Focus

- Gut health interventions to reduce total toxic load.



Foundational Therapeutic Approach: Mitigation & Clearance – Reduce Total Toxic Load



LIFESTYLE FIRST



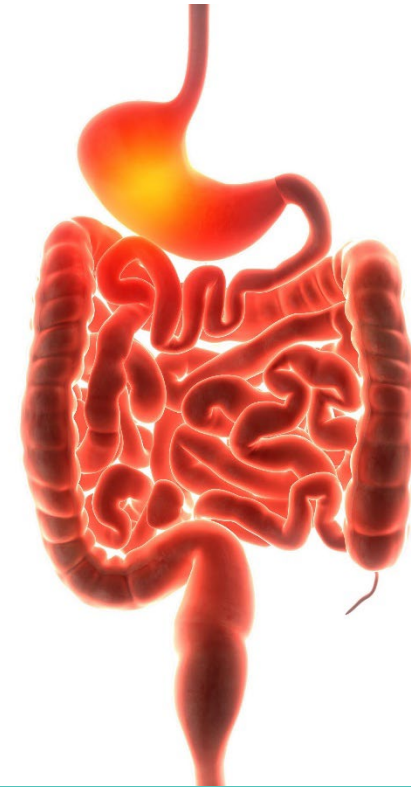
FOOD FIRST



GUT FIRST

Reducing Total Toxic Load: Functional Medicine Principle

Optimize GI Function
“Treat the Gut”



Total Toxic Load Is Determined by:

Total Exposure
(to Toxic
Substances)

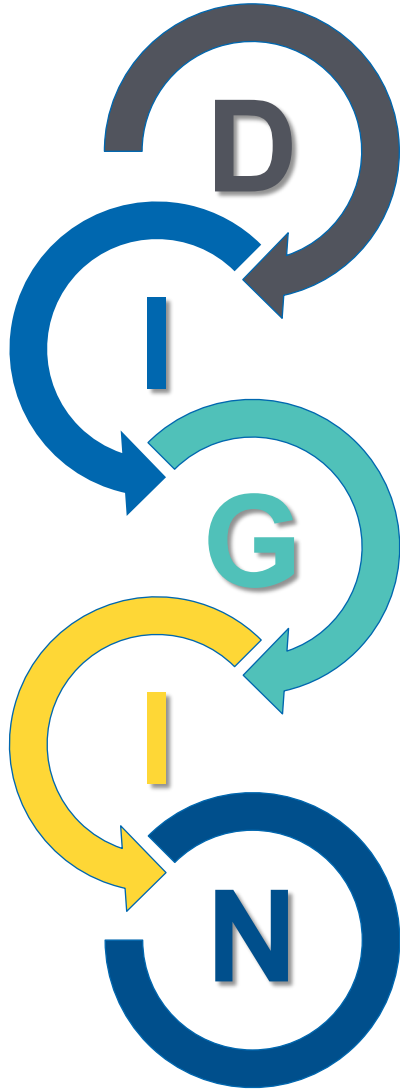


Ability to
Mitigate,
Biotransform, or
Metabolize



Clearance or
Elimination

Key Functional Roles of the Gut



Digestion/Absorption

Intestinal Permeability

Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Utilizing the 5R Bowel Program for Reducing Total Toxic Load

- **Remove** pathogens and offending foods
- **Replace** digestive enzymes and other factors
- **Repair** damaged intestinal mucosa
- **Repopulate** with prebiotics and probiotics
- **Rebalance** with lifestyle behavior changes

Clinical Focus

- Retelling the patient's story to begin co-prioritizing the importance of reducing the total toxic load.



Retelling a Patient's Story

- “You are experiencing quite a few chronic diseases, and when we see these patterns, we have to consider an elevated total toxic load as a root cause of these conditions and part of the explanation of these effects.”
- “Your life story indicates that you may have experienced significant exposures to toxic substances and that you may also have unique factors that make you more susceptible to those substances than the average person.”



Retelling a Patient's Story

- “We are certainly going to addresses the main concerns that you have about these conditions, and in addition, it would be good for us to consider how you could make some simple changes in your nutrition, your daily lifestyle routine, and your gut health to begin safely reducing your total toxic load.”
- “What do you think of this approach?”



Clinical Focus

- Summary, conclusions, and key takeaways.
- Clinical flow path.



Total Toxic Load: Key Clinical Takeaways



The effects of an elevated total toxic load are often non-specific but may be clinically suspected based on five general phenotypic patterns of clinical findings.



Susceptibility to an elevated total toxic load may be related to inherited genetic antecedents, acquired antecedents (chronic exposures and environments) and triggers (acute exposures) occurring earlier in life, as well as current undernourishment, lifestyle behaviors, and gut dysfunction (mediators).



Point-of-care questionnaires have clinical utility for identifying factors related to susceptibility, exposure, and effects of toxins and toxicants.

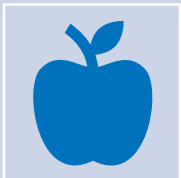
Total Toxic Load: Key Clinical Takeaways



The total toxic load is arguably more determined by factors of susceptibility and mitigation/clearance than by exposures.



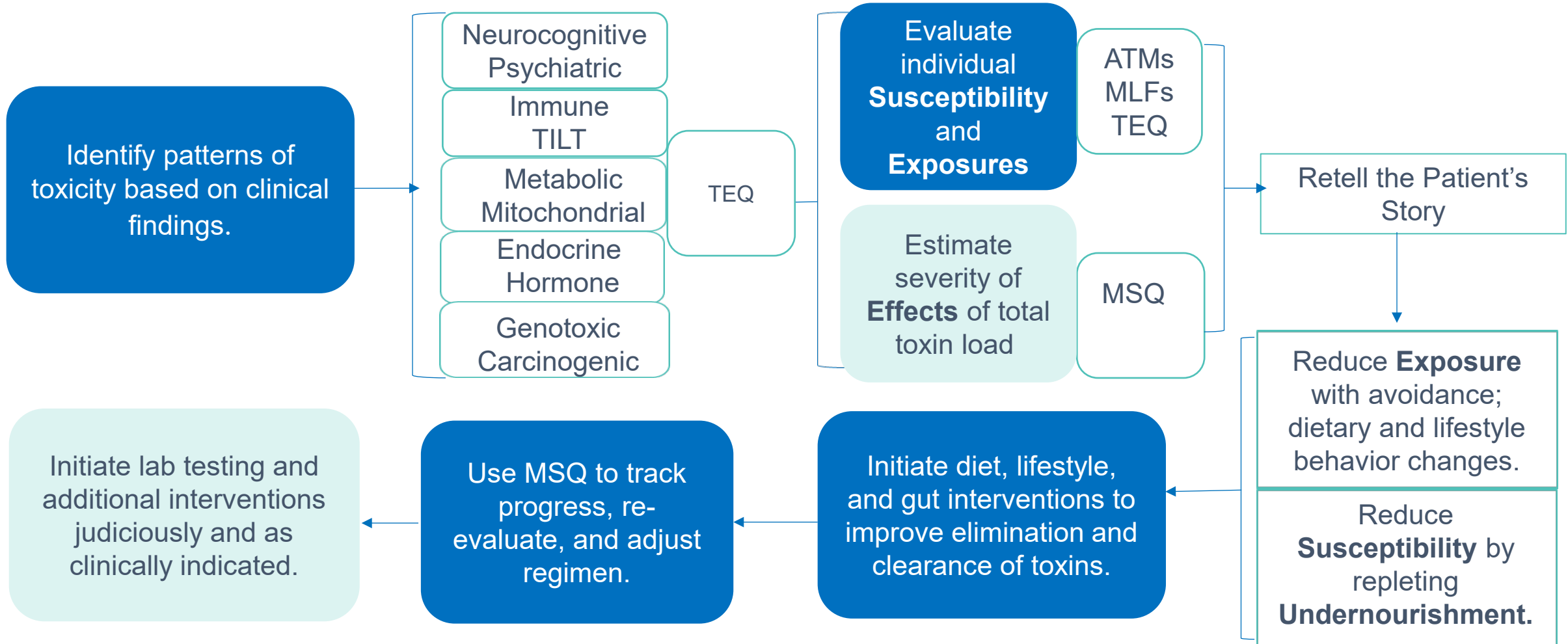
The functional goals of therapeutic interventions are to minimize exposure, balance phase I and phase II pathways, and improve mitigation/clearance of toxins.



Foundational therapeutic regimens include nutrient repletion and the optimization of lifestyle factors and gut health.

FLOW PATH:

Clinical Approach to Addressing Total Toxic Load



- TEQ: Toxin Exposure Questionnaire
- MSQ: Medical Symptoms Questionnaire

Long Description: Clinical Approach to addressing total toxic load

This graphic outlines a clinical flow path for addressing total toxic load in patients, using a step-by-step systems-based approach aligned with the GOTOIT model.

1. **Identify patterns of toxicity** based on clinical findings. Toxicity may present in various systems:
 - Neurocognitive and psychiatric
 - Immune (including T I L T: Toxicant-Induced Loss of Tolerance)
 - Metabolic and mitochondrial
 - Endocrine and hormonal
 - Genotoxic and carcinogenic
- Utilize the Toxin Exposure Questionnaire (T E Q) to assist in identifying patterns
2. **Evaluate individual susceptibility and exposures** using tools like the Toxin Exposure Questionnaire (T E Q), Antecedents, Triggers, and Mediators (A T Ms), and Modifiable Lifestyle Factors (M L Fs).
3. **Estimate the severity of effects** from the total toxin load using the Medical Symptoms Questionnaire (M S Q).
4. **Retell the patient's story** with insights gained from this evaluation.
5. Develop an **intervention plan** that includes:
 - Reducing **exposure** through avoidance, dietary and lifestyle changes.
 - Reducing **susceptibility** by correcting undernourishment.
- **Initiate diet, lifestyle, and gut interventions** to improve elimination and clearance of toxins.
- **Use the M S Q** to track progress, re-evaluate, and adjust the regimen as needed.
- **Initiate lab testing and additional interventions** judiciously based on clinical indications.

The process integrates clinical assessments with tools to evaluate exposures, symptoms, and biological function to create personalized detoxification strategies.

Environmental Health

Clinical Approach to Total Toxic Load

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE